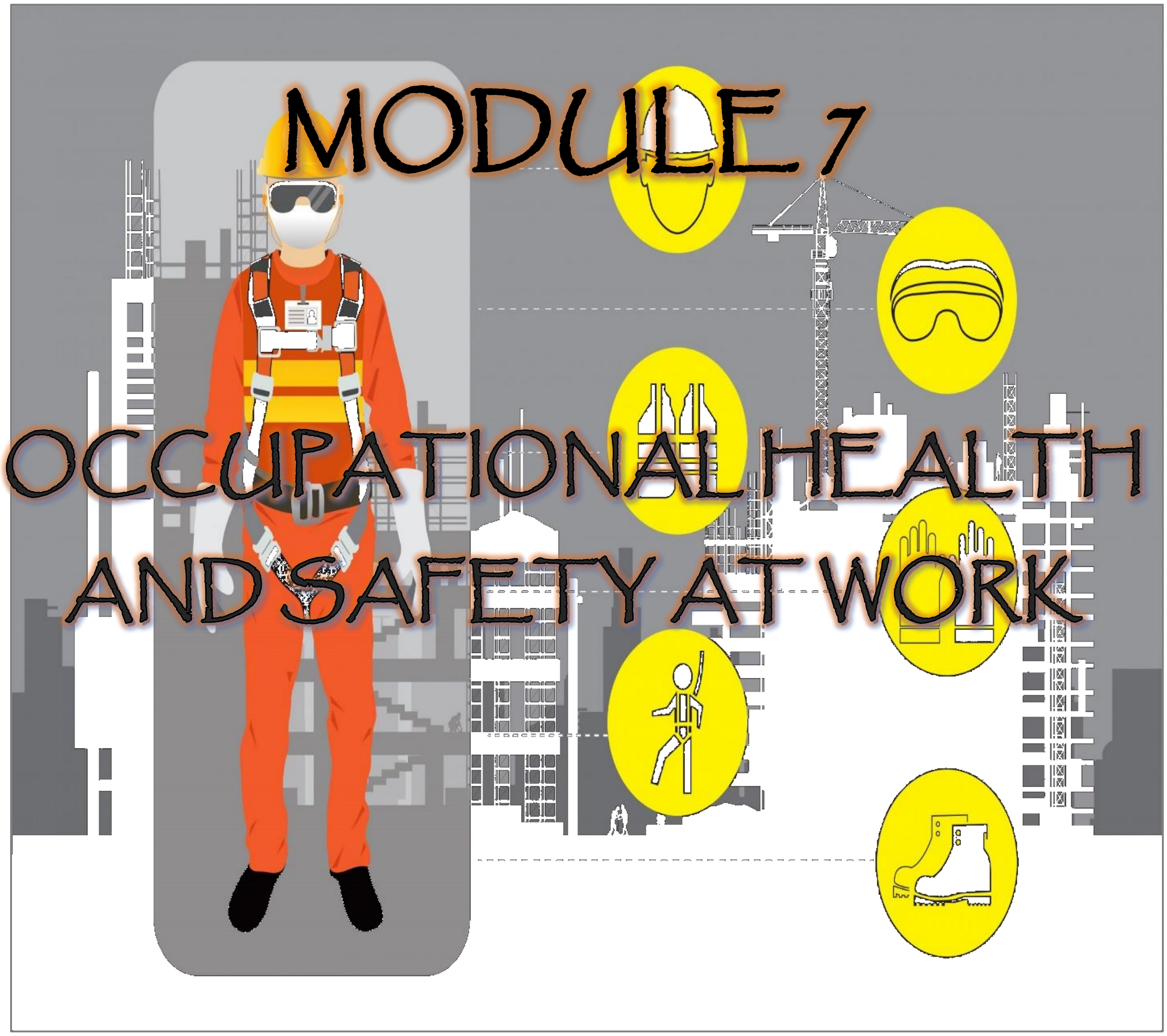
Republic of the Philippines
PARTIDO STATE UNIVERSITY
Goa, Camarines Sur

MODULE 7

OCCUPATIONAL HEALTH AND SAFETY AT WORK



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MODULE 7

Occupational Health and Safety At Work

This module aims to increase the awareness of the students for proper safety at work to lessen the risk of any accidents towards work place. Just like the expression we always heard "safety first". Because prevention towards any accident is better than cure. We only have one life so let us value it because we cannot turn back the clock once it passed.

"Every 15 seconds, 160 workers have a work-related accident. Every 15 seconds, a worker dies from a work-related accident or disease."

What is Occupational Safety and Health (OSH)?

Occupational safety and health is a discipline with a broad scope involving three major fields – Occupational Safety, Occupational Health and Industrial Hygiene.

- Occupational safety deals with understanding the causes of accidents at work and ways to prevent unsafe act and unsafe conditions in any workplace. Safety at work discusses concepts on good housekeeping, proper materials handling and storage, machine safety, electrical safety, fire prevention and control, safety inspection, and accident investigation.
- Occupational health is a broad concept which explains how the different hazards and risks at work may cause an illness and emphasizes that health programs are essential in controlling work-related and/or occupational diseases.
- Industrial hygiene discusses the identification, evaluation, and control of physical, chemical, biological and ergonomic hazards.

"In its broadest sense, OSH aims at:

- the promotion and maintenance of the highest degree of physical, mental and social well-being of workers in all occupations;
- the prevention of adverse health effects of the working conditions
- the placing and maintenance of workers in an occupational environment adapted to physical and mental needs;
- the adaptation of work to humans (and NOT the other way around).

In other words, occupational health and safety encompasses the social, mental and physical well-being of workers, that is, the "whole person".

Successful occupational health and safety practice requires the collaboration and participation of both employers and workers in health and safety programs, and involves the consideration of issues relating to occupational medicine, industrial hygiene, toxicology, education, engineering safety, ergonomics, psychology, etc.

Occupational health issues are often given less attention than occupational safety issues because the former are generally more difficult to confront. However, when health is addressed, so is safety - a healthy workplace is by definition also a safe workplace. The reverse, though, may not be true - a so-called safe workplace is not necessarily also a healthy workplace. The important point is that both health and safety issues must be addressed in every workplace."

The terms hazard and risk are often interchanged. Because you will be encountering these throughout the module it is a must that you understand the difference between them.

Hazard – a source or situation with a potential to cause harm in terms of injury, ill health, damage to property, damage to the environment or a combination of these.

Risk – a combination of the likelihood of an occurrence of a hazardous event with specified period or in specified circumstances and the severity of injury or damage to the health of people, property, environment or any combination of these caused by the event.

The hazards affecting the workplace under each major area should be detected, identified, controlled and, at best, prevented from occurring by the safety and health officer of the company. Occupational safety and health should be integrated in every step of the work process, starting from storage and use of raw materials, the manufacture of products, release of by-products, and use of various equipment and ensuring a non-hazardous or risk-free work environment.

Definition of Terms:

Occupational accident - an unexpected and unplanned occurrence, including acts of violence arising out of or in connection with work which results in one or more workers incurring a personal injury, disease or death. It can occur outside the usual workplace/premises of the establishment while the worker is on business on behalf of his/her employer, i.e., in another establishment or while on travel, transport or in road traffic.

Occupational injury - an injury which results from a work-related event or a single instantaneous exposure in the work environment (occupational accident). Where more than one person is injured in a single accident, each case of occupational injury should be counted separately. If one person is injured in more than one occupational accident during the reference period, each case of injury to that person should be counted separately. Recurrent absences due to an injury resulting from a single occupational accident should be treated as the continuation of the same case of occupational injury not as a new case.

Temporary incapacity - case where an injured person was absent from work for at least one day, excluding the day of the accident, and 1) was able to perform again the normal duties of the job or position occupied at the time of the occupational accident or 2) will be able to perform the same job but his/her total absence from work is expected not to exceed a year starting the day after the accident, or 3) did not return to the same job but the reason for changing the job is not related to his/her inability to perform the job at the time of the occupational accident.

Permanent incapacity - case where an injured person was absent from work for at least one day, excluding the day of the accident, and 1) was never able to perform again the normal duties of the job or position occupied at the time of the occupational accident, or 2) will be able to perform the same job but his/her total absence from work is expected to exceed a year starting the day after the accident.

Fatal case - case where a person is fatally injured as a result of occupational accident whether death occurs immediately after the accident or within the same reference year as the accident.

Frequency Rate (FR) – refers to cases of occupational injuries with workdays lost per 1,000,000 employee-hours of exposure.

Incidence Rate (IR) – refers to cases of occupational injuries with workdays lost per 1,000 workers.

Severity Rate (SR) – refers to workdays lost of cases of occupational injuries resulting to temporary incapacity per 1,000,000 employee-hours of exposure.

Average Workdays Lost – refer to workdays lost for every case of occupational injury resulting to temporary incapacity.

Unsafe / Unhealthy Acts and Conditions

Accidents
An accident is an unexpected, unforeseen, unplanned and unwanted occurrence or event that causes damage or loss of materials or properties, injury or death.

Common types of accidents:

- fall from height and fall from the same level (slips and trips)
- struck against rigid structure, sharp or rough objects
- struck by falling objects
- caught in, on or in between objects
- electrocution
- fire

Costs of accidents

Corollary to accidents are costs that companies have to bear whether directly or indirectly. The cost of accidents can be best explained by the Iceberg Theory. Once an accident happens, money has to be spent for medical expenses of the injured worker/workers, insurance premiums and, in some cases, for penalty and litigation expenses. Companies also spend huge amounts to replace damaged equipment and wasted raw materials. These are what we consider as the direct costs of accidents. But these are just the tip of the iceberg.

The larger and more dangerous part of the iceberg however is the part that lies beneath the water. This represents the indirect costs of an accident which have a more damaging impact to the worker, their families, the company and the community in general. Indirect costs include:

1. Lost or lesser productivity of the injured – workers lose their efficiency and income due to work interruption on the day of the injury.
2. Loss of productivity among other employees due to work stoppage when assisting the injured worker, inspection or merely out of curiosity. The psychological impact of the accident reduces the workers' productivity.
3. Loss of productivity among supervisors because instead of focusing on managing people and the work flow, they spend their time assisting the injured, investigating the accident and preparing inspection reports.
4. Hiring and training replacement workers
5. Downtime due to equipment damage

Apart from these are humane aspects of accidents such as sorrow due to loss, hardships and inconveniences, physical pain and discomfort and psychological problems.

Accident causation

Are these phrases familiar to you?

– "Oras na niya"
– "Malas niya lang"
– "Tanga kasi"
– "Kasama sa trabaho"

People usually utter the abovementioned phrases or statements when someone gets injured or dies in an accident. However, these are not the real causes of accidents but mere excuses of people who do not understand the concepts of occupational safety and health. Accidents are primarily caused by unsafe and unhealthy acts and conditions.

Unsafe/unhealthy Act: the American National Standards Institute (ANSI) defines this as "any human action that violates a commonly accepted safe work procedure or standard operating procedure." This is an act done by a worker that does not conform or departs from an established standard, rules or policy. These often happen when a worker has **improper attitudes, physical limitations or lacks knowledge or skills**.

Examples of unsafe acts include: horse playing, smoking in non-smoking areas, using substandard/defective tools, non - wearing of goggles/gloves, driving without license, reporting to work under the influence of liquor or drugs, and improper storage of paints and hazardous chemicals among others.

Unsafe/unhealthy Condition: ANSI defines this as the physical or chemical property of a material, machine or the environment which could possibly cause injury to people, damage to property, disrupt operations in a plant or office or other forms of losses. These conditions could be guarded or prevented.

Examples of unsafe conditions include: slippery and wet floors, dusty work area, congested plant lay-out, octopus wiring, scattered objects on the floor/work area, poor storage system, protruding nails and sharp objects, unguarded rotating machines/equipment, etc.

Can accidents be prevented?

Herbert William Heinrich, an American industrial safety pioneer who worked as an Assistant Superintendent of the Engineering and Inspection Division of Travelers Insurance Company, did a study on the insurance claims. After reviewing thousands of accident reports completed by supervisors, who generally blamed workers for causing accidents without conducting detailed investigations into the root causes, Heinrich found out that 98% of workplace accidents are preventable and only 2% are non-preventable. Of the 98% preventable accidents, 88% is due to unsafe/unhealthy acts or "man failure" and 10% is due to unsafe/unhealthy conditions. This study explains the rationale for focusing interventions on changing the behaviors and attitudes of workers and management towards safety and health.

How do you prevent yourself from performing unsafe/unhealthy acts that will cause unsafe/unhealthy conditions at work?
It is important to raise everybody's consciousness to such a degree that we all begin to realize that our actions affect other people in the workplace, even if these appear to have nothing to do with them. If you agree that we are part of the problem, then, probably we can be part of the solution, too.

Occupational Safety

HOUSEKEEPING
This module aims to introduce you to the importance of good housekeeping in preventing most common accidents in the workplace (we also think it will be good to implement in your homes and schools).
The 5S, a Japanese concept that aims to optimize time for production, is a very practical, simple and proven approach to improving housekeeping in the workplace. Housekeeping is important because it lessens accidents and related injuries and illnesses. It therefore improves productivity and minimizes direct/indirect costs of accidents/illnesses. Housekeeping means putting everything in its proper place. It is everybody's business to observe it in the workplace.

Defining Housekeeping

Let us begin by showing you what housekeeping is not: It is shown when your surroundings have:
- cluttered and poorly arranged areas
- untidy piling of materials
- improperly piled-on materials that results to damaging other materials
- items no longer needed
- blocked aisles and passageways
- materials stuffed in corners and out-of-the-way places
- materials getting rusty and dirty from non-use
- excessive quantities of items
- overcrowded storage areas and shelves
- overflowing bins and containers
- broken containers and damaged materials

Do you agree with this? Housekeeping is avoiding all of the above and many more. Now instead of just being crabby and complaining about poor housekeeping, why don't we see how we can instill and implement good housekeeping in our workplace? Look at the two pictures below. Do you know about with these seven wastes and how we can eliminate them? You got it! Through good housekeeping!

What is 5S?

5S is a systematized approach to:

- organizing work areas
- keeping rules and standards
- maintaining discipline

5S utilizes:

- workplace organization
- work simplification techniques

5S practice…

- develops positive attitude among workers
- cultivates an environment of efficiency, effectiveness and economy

5S Philosophy

- Productivity comes from the elimination of waste
- It is necessary to attack the root cause of a problem, not just symptoms
- Participation of everybody is required
- To acknowledge that the human being is not infallible

5S Terms:

1. Seiri/Sort/Suriin – is the first S which means sorting out unnecessary items and discarding them.
    - Make the work easy by eliminating obstacles
    - Eliminate the need to take care of unnecessary items
    - Provide no chance of being disturbed with unnecessary items
    - Prevent faulty operation caused by unnecessary items.

2. Seiton/Systematize/ Sinupin – is the second S which means we need to organize things

7 Seiton Principles:

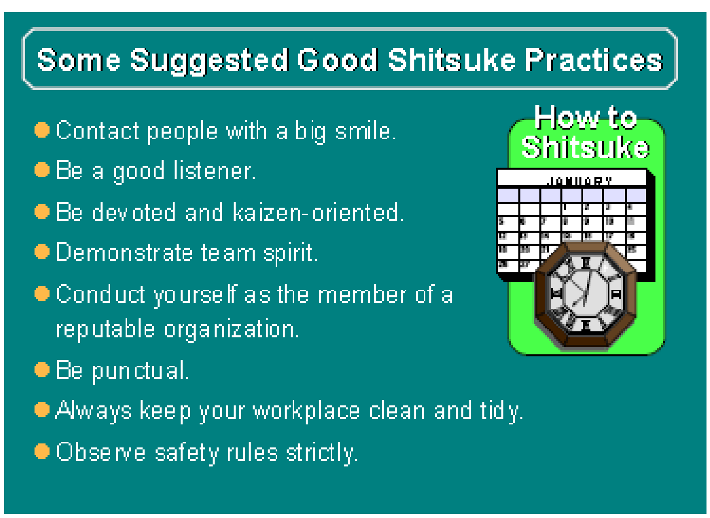
- Follow the first-in-first-out (FIFO) method for storing items
- Assign each item a dedicated location.
- All items and their locations should be indicated by a systematic labeling
- Place items so that they are visible to minimize search time
- Place items so they can be reached or handled easily.
- Separate exclusive tools from common ones.
- Place frequently used tools near the user.

3. Seiso/Sweep/Simutin – is the third S which means we have to sanitize or clean our workplace.

- Keep environmental condition as clean as the level necessary for the products
- Prevent deterioration of machinery and equipment and make checking of abnormalities easy
- Keep workplace safe and work easy

4. Seiketsu/Standardize/Siguruhin – is the fourth S which means we have to standardize what we are doing.

5. Shitsuke/Self- Discipline/Sariling kusa – is the fifth and last S which means we have to do this process without prodding.



Good housekeeping is needed for quality improvement

By this we lessen rejects/losses. If the workplace is in order, it is easy to do the job. An easier job, having no defects, continuous production and an orderly workplace is akin to work improvement. And now that ISO Certification is the trend, the impression of a company to the community is very important. A company that follows good housekeeping principles will surely be recognized as a provider of quality service and products.

Steps in implementing 5S

Step 1: Preparations

a. Understanding 5S concepts and benefits by the CEO
b. CEO's visit to the 5S model companies
c. CEO's commitment to 5S implementation
d. Organize 5S working Committee
e. 5S facilitators
f. Train facilitators and practitioners

Step 2: Management's official announcement

a. CEO officially announces implementation of 5S program
b. CEO explains the objectives of 5S to all colleagues
c. Publicize 5S organizational chart and lay-out
d. Work out various promotional tools

5S CORE GROUP
ROLES AND RESPONSIBILITIES

- **PLAN**
  - Situation Appraisal
  - Setting Benchmarks or Targets
  - Implementation Plan
- **DO**
  - Announcements
  - Education
  - Akafuda
  - Big Seiso
  - Seiso Inspection
  - Seiton Campaign
  - Special 5S Projects
- **CHECK**
  - 5S Audit
  - Documentation of Targets
  - Review Targets
- **ACT**
  - Corrective Measures
  - Revise Plans
  - Difficulties Encountered

Management's Role

• Providing adequate equipment
• Including housekeeping in the planning of all operations
• Including maintenance of good housekeeping as part of individual's job responsibility
• Providing clean-up schedule and personnel
• Maintaining executive supervisory and interest

Supervisor's Role

• Maintaining constant check on housekeeping conditions
• Seeing that employees maintain good housekeeping
• Having unusual situations corrected or cleaned up immediately
• Planning for orderliness in all operations
• Issuing definite instructions to employees

• Insisting on clean-up after every job

Worker's Role

• Follow housekeeping procedures
• Maintain an orderly workplace
• Report to supervisors any unsafe condition

Step 3: Big clean-up day

a. Organize a big clean-up day after 5S implementation announcement by CEO
b. Divide company premises into small areas and assign a small group of people for each area
c. Provide enough cleaning tools and materials
d. This big cleaning must include public areas such as gardens, canteen and car park
e. Everybody must participate in this big cleaning day

Step 4: Initial seiri

a. Establish disposal standards for unnecessary items
b. Apply "Disposal Notices" to all questionable items
c. Carefully examine responses to disposal notices
d. Dispose unnecessary items according to disposal standards
e. A company-wide seiri should be planned and practiced annually

Daily Seiso and Seiton activities

a. Identify areas for improvement and work out a priority listing by colleagues
b. Select untidy, inconvenient and unsafe areas
c. Set each activity for 3-6 months
d. Organize presentations by small groups
e. Standardize good 5S practices visibly
f. Motivate colleagues for creative improvements

Hard 5S – refers to all facets of the work environment

a. Furniture – tables, shelves, drawers
b. Equipment – computers, projector, fax, copier
c. Lay-out of desk and equipment

Soft 5S

a. Office policies and procedures
b. Dress code
c. Sharing of responsibilities, telephone etiquette

5S Office guidelines

Desks

- Do not place anything under your desk (Seiton)
- Dispose of unnecessary items in your drawers (Seiri)
- Arrange items in your desk drawers neatly for easy retrieval (Seiton)
- Do not pile up documents on your desk top (Seiton)
- Wipe your desktop every morning
- Do not leave unnecessary things on your desk top when you go home (Seiton)

Office machines

- Clean office machines and equipment regularly (Seiso)
- Set electric cables neatly for safety and good appearance (Seiton)
- inspect machines regularly and take action for required servicing (Shitsuke)

Toilets

- Flush after use (Seiketsu)
- Wash hands after using the toilet (Seiketsu)
- Clean up toilet and wash basin everyday (Seiso/Seiketsu)
- Replenish toilet paper, soaps and paper towels (Seiton/Seiketsu)
- All users should always try to keep toilets clean and tidy (Shitsuke)
- Check exhaust fans regularly for effective function (Seiso)

Canteen

- Do not leave unnecessary things on the dining table (Seiton/Seiketsu)
- Tuck chairs properly after use (Seiton)
- Return chairs and tables to their original location when used for meetings or functions (Seiton)
- Put away all cups and plates after each meal (Seiso/Seiketsu)
- Clean up tables immediately after each meal (Seiso/Seiketsu)

Hallways

- Do not smoke while walking in the hallways (Shitsuke)
- Do not place anything in the hallways without permission (Seiri/Seiton/Shitsuke)
- Pick-up and dispose any waste in the hallway (Seiketsu/Shitsuke)
- Avoid talking loudly along hallways (Shitsuke)

Notice Boards

- Ensure that outdates notices are removed (Seiketsu)
- Ensure that all information are updated regularly (Seiri)
- Items should be neatly aligned and properly secured (Seiton)
- Pins must be readily available (Seiton)
- Check that the location of notice boards are appropriate (Seiton)

Visual Control - a technique that enables people to make the rules easy to follow, differentiate normal and abnormal situations and act accordingly, with the use of visual aids.

Pointers in making visual control

a. Should be easy to see from a distance
b. Should be properly and strategically located
c. Should be easy to follow
d. Should facilitate distinction of what is right and what is wrong

Step 5: Periodic 5S audits

a. Establish 5S evaluation and incentive plan
b. Conduct 5S evaluation and inspection regularly
c. Organize 5S inter-department competition
d. Periodically award groups and individuals
e. Organize study tours to other companies
f. Organize 5S inter-company competition

Purpose of 5S audit

a. Turn PDCA Cycle (Plan-Do-Check-Act)
b. Analyze the results of actual implementation in the workplace
c. Give support and guidance to the members of each unit
d. Dissemination of good practices
e. Regular audit sustains the program

Key points in the implementation of 5S

1. Start small, easy and proceed slowly but steadily
2. Start with the most suitable “S”
3. Only one or two “S” are enough for the initial practice
4. Set simple, easily achievable and step by step targets
5. Everyone’s participation is important
6. Management should take leadership of 5S movement
7. Record improvements for comparison
8. Devise schemes to stimulate awareness and stimulate enthusiasm

Factors leading to the success of 5S

a. Strong sponsorship and leadership of CEO
b. Active promoter/5S committee
c. Good launching activity
d. Regular audits
e. Good documentation
f. Visits by external consultants
g. Competition

Factors that hinder the success of 5S

a. Project sponsor is not the decision maker
b. Organizational policies
c. Lack of experience in undertaking cross-functional activities
d. Lack of top management support
e. Implementation carried out through orders from the management
f. Implementation done by task forces
g. 5S treated as a project
h. Emphasis on immediate results

Why should we pay attention to housekeeping at work?

Effective housekeeping can eliminate some workplace hazards and help get a job done safely and properly. Poor housekeeping can frequently contribute to accidents by hiding hazards that cause injuries. If the sight of paper, debris, clutter and spills is accepted as normal, then other more serious health and safety hazards may be taken for granted.

Housekeeping is not just cleanliness. It includes keeping work areas neat and orderly; maintaining halls and floors free of slip and trip hazards; and removing of waste materials (e.g., paper, cardboard) and other fire hazards from work areas. It also requires paying attention to important details such as the layout of the whole workplace, aisle marking, the adequacy of storage facilities, and maintenance. Good housekeeping is also a basic part of accident and fire prevention.

Effective housekeeping is an ongoing operation: it is not a hit-and-miss cleanup done occasionally. Periodic "panic" cleanups are costly and ineffective in reducing accidents.

What is the purpose of workplace housekeeping?

Poor housekeeping can be a cause of accidents, such as:

- tripping over loose objects on floors, stairs and platforms
- being hit by falling objects
- slipping on greasy, wet or dirty surfaces
- striking against projecting, poorly stacked items or misplaced material
- cutting, puncturing, or tearing the skin of hands or other parts of the body on projecting nails, wire or steel strapping

To avoid these hazards, a workplace must "maintain" order throughout a workday. Although this effort requires a great deal of management and planning, the benefits are many.

What are some benefits of good housekeeping practices?

Effective housekeeping results in:

- reduced handling to ease the flow of materials
- fewer tripping and slipping accidents in clutter-free and spill-free work areas
- decreased fire hazards
- lower worker exposures to hazardous substances (e.g. dusts, vapours)
- better control of tools and materials, including inventory and supplies
- more efficient equipment cleanup and maintenance
- better hygienic conditions leading to improved health
- more effective use of space
- reduced property damage by improving preventive maintenance
- less janitorial work
- improved morale
- improved productivity (tools and materials will be easy to find)

What is an example of a workplace housekeeping checklist for construction sites?

DO:

- Gather up and remove debris to keep the work site orderly.
- Plan for the adequate disposal of scrap, waste and surplus materials.
- Keep the work area and all equipment tidy. Designate areas for waste materials and provide containers.
- Keep stairways, passageways, ladders, scaffold and gangways free of material, supplies and obstructions.
- Secure loose or light material that is stored on roofs or on open floors.
- Keep materials at least 2m (5 ft.) from openings, roof edges, excavations or trenches.
- Remove or bend over nails protruding from lumber.
- Keep hoses, power cords, welding leads, etc. from lying in heavily travelled walkways or areas.
- Ensure structural openings are covered/protected adequately (e.g. sumps, shafts, floor openings, etc.)

DO NOT:

- Do not permit rubbish to fall freely from any level of the project. Use chutes or other approved devices to materials.
- Do not throw tools or other materials.
- Do not raise or lower any tool or equipment by its own cable or supply hose.

Flammable/Explosive Materials

- Store flammable or explosive materials such as gasoline, oil and cleaning agents apart from other materials.
- Keep flammable and explosive materials in proper containers with contents clearly marked.
- Dispose of greasy, oily rags and other flammable materials in approved containers.
- Store full barrels in an upright position.

- Keep gasoline and oil barrels on a barrel rack.
- Store empty barrels separately.
- Post signs prohibiting smoking, open flames and other ignition sources in areas where flammable and explosive materials are stored or used.
- Store and chain all compressed gas cylinders in an upright position.
- Mark empty cylinders with the letters "mt," and store them separately from full or partially full cylinders.
- Ventilate all storage areas properly.
- Ensure that all electric fixtures and switches are explosion-proof where flammable materials are stored.
- Use grounding straps equipped with clamps on containers to prevent static electricity buildup.
- Provide the appropriate fire extinguishers for the materials found on-site. Keep fire extinguisher stations clear and accessible.

What is an example of a Housekeeping Inspection Checklist?
Use the following checklist as a general workplace guide.

Floors and Other Areas
- Are floors clean and clear of waste?
- Are signs posted to warn of wet floors?
- Are floors in good condition?
- Are there holes, worn or loose planks or carpet sticking up?
- Is anti-slip flooring used where spills, moisture or grease are likely?
- Are there protruding objects such as nails, sharp corners, open cabinet drawers, trailing electrical wires?
- Are personal items, such as clothing and lunch boxes, in assigned lockers or storage areas?
- Is the work area congested?
- Are floors well-drained?

Aisles and Stairways
- Are aisles unobstructed and clearly marked?
- Are mirrors installed at blind corners?
- Are aisles wide enough to accommodate workers and equipment comfortably?
- Are safe loading practices used with hand and power trucks, skids, or pallets?
- Is the workplace lighting adequate? Are stairs well lit?
- Are stairs covered with an anti-slip tread? Are faulty stair treads repaired?

Spill Control
- Are all spills wiped up quickly?
- Are procedures followed as indicated on the material safety data sheet?
- Are spill absorbents used for greasy, oily, flammable or toxic materials?
- Are used rags and absorbents disposed of promptly and safely?
- Is a spill area surrounded by a barrier to prevent a spill from spreading?

Equipment and Machinery Maintenance
- Is equipment in good working order, with all necessary guards or safety features operational or in place?
- Is equipment damaged or outdated?
- Are tools and machinery inspected regularly for wear or leaks?
- Is equipment repaired promptly?
- Are drip pans or absorbent materials used if leaks cannot be stopped at the source?
- Is a machine that splashes oil fitted with a screen or splash guard?
- Are machines and tools cleaned regularly?

Waste Disposal

- Are there adequate numbers of containers?
- Are there separate and approved containers for toxic and flammable waste?
- Are waste containers located where the waste is produced?
- Are waste containers emptied regularly?
- Are toxic and flammable waste chemicals handled properly?

Storage

- Are storage areas safe and accessible?
- Is material stacked securely, blocked or interlocked if possible?
- Are materials stored in areas that do not obstruct stairs, fire escapes, exits or firefighting equipment?
- Are materials stored in areas that do not interfere with workers or the flow of materials?
- Are bins or racks provided where material cannot be piled?
- Are all storage areas clearly marked?
- Do workers understand material storage and handling procedures?

Fire Prevention

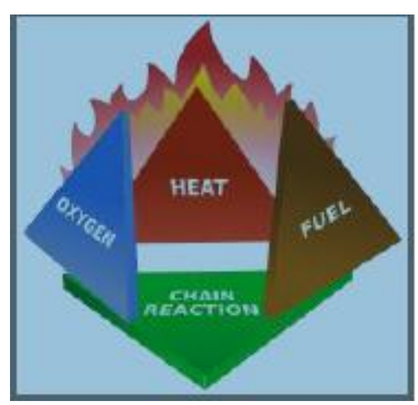
- Are combustible and flammable materials present only in the quantities needed for the job at hand?
- Are combustible and flammable materials kept in safety cans during use?
- Are hazardous materials stored in approved containers and away from ignition sources?
- Are sprinkler heads clear of stored material?
- Are fire extinguishers inspected and located along commonly travelled routes, and close to possible ignition sources?
- Are oily or greasy rags placed in metal containers and disposed of regularly?

Fire Safety

Fire is a chemical reaction between a flammable or combustible material and oxygen. This process converts the flammable or combustible materials and oxygen into energy. Other by-products of fire include light, smoke and other gases. Many of these gases such as carbon monoxide, carbon dioxide, hydrogen bromide, hydrogen cyanide, hydrogen sulfide, sulfur dioxide, nitrogen dioxide, etc. are toxic to humans.

The Fire Triangle and the Fire Tetrahedron

The fire triangle and the pyramid of fire illustrate the elements necessary for fire to start and the methods of extinguishment. Each side represents an essential ingredient for fire. The three elements are Fuel, Oxygen and Heat. When a fire starts, a fourth element, which is the chemical reaction itself, is necessary for flame propagation. The four-sided figure is called the **Fire Tetrahedron**.



Fuel: Any material that will burn is classified as fuel.

Flammable substance – is a substance having a flashpoint below 100 °F (37.8 °C) and vapor pressure not exceeding 20 psia at 100 °F. Examples of flammable substances with their flashpoint are shown below:

	Flashpoint	
Flammable substances	**°F**	**°C**
Gasoline	-45	-42.8
Ether	-49	-45
Acetone	0	-17.8
Alcohol	55	12.8

Combustible substance
– is a substance having a flashpoint at or above 100 °F (37.8 °C).

	Flashpoint	
Combustible substances	**°F**	**°C**
Fuel Oil	100	37.8
Kerosene	100	37.8
Quenching Oil	365	185.0
Mineral Oil	380	193.3

Flashpoint - the lowest temperature at which fuel begins to give off flammable vapors and form an ignitable mixture in air.

Which is more dangerous: a substance with low flashpoint or a substance with a high flashpoint?
Answer: The lower the flashpoint, the more dangerous a substance is.

Oxygen
From our definition of FIRE, we need oxygen which combines with fuel while burning. Normally, the air has 21% oxygen and 78% while nitrogen.

Heat – completes the chemistry of fire
Even if found together, fuel and oxygen will not burn. An example is a piece of paper. This fuel is exposed to oxygen in the air but will not burn. Why? Because we need to introduce the third element which is Heat.
It is when we heat up the piece of paper sufficiently that it will start to burn.

What we form with these three elements is called the Fire Triangle. This model shows us that to have fire we need three elements. And if these elements are combined at the right proportion, we will have fire.

Classification of fires
There are four classes of fires, categorized according to the kind of material that is burning. For the first three classes of fires, there are two sets of color-coded icons commonly used. One or both kinds of icons appear on most fire extinguishers to indicate the kinds of fire against which the unit is intended to be used.

Class A fires are those fueled by materials that, when they burn, leave a residue in the form of ash, such as paper, wood, cloth, rubber, and certain plastics.

Class B fires involve flammable liquids and gasses, such as gasoline, paint thinner, kitchen grease, propane, and acetylene.

Class C fires - Fires that involve energized electrical wiring or equipment (motors, computers, panel boxes). Note that if the electricity to the equipment is cut, a Class C fire becomes one of the other three types of fires.

Class D fires involve combustible metals such as magnesium, sodium, titanium, and certain organometallic compounds such as alkyl lithium and Grignard reagents.

Class K fires Fire that involves combustible cooking fuels such as vegetable or animal oils and fats.

COMMON CAUSES OF FIRES:

Electricity
Hazards of electricity involve electrocution and fire. Usually, fire is caused by overheating, arcs and sparks.
Overheating happens when there is overloading of system, short circuit and poor insulation. These are caused by improper wiring connection/practice, tampering with safety devices such as fuse and circuit breakers, and old and poorly maintained electrical installation.

Control
- conduct regular inspection and maintenance of electrical installation
- employ trained and licensed electrician
- follow Philippine Electrical Code and Occupational Safety and Health Standards

Arcs and sparks normally happen when one opens or closes a circuit. The danger arises when arcs and sparks occur in a flammable or explosive atmosphere which could result to explosion. To control arcs and sparks, use explosion proof equipment or intrinsically safe devices.

Mechanical heat
Heated surfaces on furnaces, flues, heating devices and light bulbs can cause fires if flammable or combustible materials are close enough to absorb sufficient heat to cause combustion. Care should be taken to ensure that all such devices are properly installed, especially with respect to clearance and barrier materials.

Friction sparks
Friction generates heat. Excessive heat generated by friction causes a very high percentage of industrial fires. Fire usually results from:

- ➢ overheated power-transmission bearings and shafting from poor lubrication and excessive dust
- ➢ jamming of work material during production
- ➢ incorrect tension adjustment of belt-driven machinery. If the belt is too tight or too loose, excessive friction could develop

Control

- ➢ Preventive maintenance program to keep bearings well oiled and do not run hot. And keep accumulation of flammable dust or lint on them to a minimum.
- ➢ Keep oil holes of bearings covered to prevent dust and gritty substances from entering the bearings.

Open flames
Carelessly discarded cigarettes, pipe embers, and cigars are a major source of fire. Prohibit smoking, especially in woodworking shops, textile mills, four mills, grain elevators, and places where flammable liquids or combustible products are manufactured, stored or used.

Control

- ➢ providing a "No Smoking Area" at specified times where supervision can be maintained.
- ➢ marking areas where exposure is severe with conspicuous “No Smoking” signs, prohibiting employees from even carrying matches, lighters and smoking material of any kind

Spontaneous heat (auto-ignition)

Spontaneous ignition results from a chemical reaction where there is a slow generation of heat from oxidation of organic compounds that, under certain conditions, is accelerated until the ignition temperature of the fuel is reached.

It is a condition usually found only in quantities of bulk material packed loosely enough, with large amount of surface to be exposed to oxidation, yet without adequate air circulation to dissipate heat.

Welding and cutting sparks

Hazardous sparks such as globules of molten, burning metal or hot slag are produced by both welding and cutting operations. Sparks from cutting, particularly oxy-fuel gas cutting, are generally more hazardous than those from welding because the sparks are more numerous and travel greater distances.

Control time for welding and cutting:

- Move combustibles a safe distance away - 35 ft. horizontally or Move work to a safe distance
- Protect the exposed combustibles with suitable fire resistant guards and provide a trained fire watcher with extinguishing equipment readily available
- Cover openings in walls, floors or ducts should be if within 35 ft of the work.
- Implement "Hot Work Permit System"

Highly flammable or combustible materials

Take care that the following materials are not stored with machinery or near any type of electrical or heat source. Highly flammable materials may include:

- Hay and straw
- Bedding material (especially sawdust and shredded newspaper)
- Cobwebs, dust, and grain dust
- Horse blankets
- Paint
- Fertilizer
- Pesticides and herbicides

Accelerants

Accelerants are substances that increase the speed at which a fire spreads. All accelerants are highly flammable or combustible, but not all highly flammable or combustible materials are accelerants. Accelerants must be stored in approved containers and properly labeled as such (plastic milk bottles do not qualify as approved containers for storing chemicals). An updated list of all chemicals in the workplace should be maintained. The list should include the name of the chemical, date purchased, the quantity of the chemical, and the place of storage on the farm. This list should be kept in a safe, handy place such as an office (not in the building where the products are stored). In case of a fire, the list should be given to the fireperson in charge to aid the fire department in knowing what potential toxic fumes or explosions may result and how best to contain the situation.

Common accelerants include:

- Gasoline
- Kerosene
- Oil
- Aerosol cans

Ignition sources –

An ignition source is something that can cause an accelerant or flammable material to ignite or smolder.

Examples of ignition sources are:

- Cigarettes and matches
- Sparks from welding machines and machinery (trucks, tractors, mowers)
- Motors

- Heaters
- Electrical appliances
- Electrical fixtures and wires
- Batteries
- Chemicals which may react with each other or with water or dampness

Provide for early detection of fire

Except for explosions, most fires start out as small ones. At the initial stage, extinguishing a fire seldom presents much of a problem. Once the fire begins to gain headway, it may develop into conflagration of disastrous proportions. Fire can be more easily controlled if detected early. It is critical that fire be extinguished in the first five minutes.

Detection serves to:

- warn the fire brigade to start extinguishing procedure
- warn occupants to escape

Means of detection include:

- human observer
- automatic sprinklers
- smoke, flame and heat detectors

a. Smoke detectors

- Monitor changes within the area
- Provide early warning
- Changing stages in the development of fire
- When smoke is produced

b. Heat detectors

- Fixed temperature types – which responds when the detection element reaches a predetermined temperature
- Rate-of-rise temperature – which respond to an increase in heat at a rate greater than some predetermined value.

c. Flame detectors

- Infra-red – sensing elements responsive to radiant energy outside the range of human vision; useful in detecting fire in large areas, e.g. storage areas
- Ultra-violet – sensing elements responsive to radiant energy outside the range of human vision

Prevent the spread of fire

Once a fire is discovered, it is of prime importance to confine it to the smallest area possible - that is, to prevent its spread. This can be accomplished by details of construction and by safe practices, but neither is sufficient alone. An understanding of the means by which heat is transmitted will be of value in taking the necessary steps to prevent the spread of fire.

These are the **three (3) methods of heat transfer** and how it can be controlled

- **Conduction** is the transfer of heat from molecule to molecule. Thermal conductivity is important in terms of fire spread. A steel girder passing through an otherwise fireproof wall may cause fire spread by conducted heat.
- **Convection** is caused by movement of heated gasses produced by any burning material or by heated air rising to the upper limits of the space in which it is contained. During a fire in a building convection currents convey combustion gases up through stairways or lift shafts, spreading the fire to other parts of the building.

- **Radiation** is the transfer of heat in straight rays.

Control
Barriers are one means of control that will limit the area of a fire or at least retard its spread. Examples are: firewalls, fire doors, shutters or louvers, fire stops, baffles, fire dampers, fire windows, parapets, dikes and enclosures of vertical openings
Provide for prompt extinguishment
In providing for prompt extinguishment, the two categories of fire extinguishers should be kept in mind – permanent or “built-in” extinguishers and portable fire extinguishers.

Permanent or "built-in" fire extinguishers
Examples include:
- Standpipe and hose
- Automatic sprinkler system
- Fire hydrant
- Fire pump
- Fire truck
- Automatic extinguishing system

Portable extinguishers
These are used extensively to lessen the danger from fire. After such a system is installed, its proper maintenance and regular inspection is suggested to ensure its usefulness when needed.
Portable fire extinguishers are also called first-aid fire extinguishers since they are intended to be used for incipient fires. They contain a limited supply of an extinguishing medium. These appliances are designed for use on fires of specific classes.

Requirements for effective use of fire extinguishers:
1. **Of the approved type**
- must have a seal of PS mark for locally made, and UL mark for imported ones
2. **The right type for each class of fire that may occur in the area**
3. **In sufficient quantity**
- The number of fire extinguishers must be computed according to the floor area, the degree of hazard of fire that may occur and the class of fire.
4. **Located where they are easily accessible for immediate use and the location is kept accessible and clearly identified.**

In the absence of modern fire extinguishers, the following can be used to stop fire in its initial stage.
- For A fire - water is the best.
- For B fire - a metal cover, wet sack, towel, cloth, or blanket will do. Sand and soil are very useful
- For C fire - the main switch is the first consideration. Pull it down to cut off the current. What is useful on A & B can also be useful here.

Remember the **PASS** - word
P - pull the pin
A - aim low
S - squeeze the lever above the handle
S - sweep from side to side

General fire safety precautions

- Smoking should never be permitted in any storage area, tack room or lounge. “No-Smoking” signs should be posted in these areas and at all exterior entrances. Butt cans should be provided as an incentive to extinguish all cigarettes.
- Exit doors should be clearly marked.
- Aisles should be raked or swept clean at all times. Vacuum up cobwebs and dust regularly. Wipe dust/dirt off light fixtures, outlet covers, switches and panel boxes.
- Weeds, twigs, and other trash should be kept mowed or picked up from around the outside of the building.
- Paper storage should not be near lights, fans, electrical boxes, heaters or outlets.
- Flammable substances should be kept elsewhere outside the building.
- Vehicles and machinery should be stored in a separate building.
- A fire hose and buckets should be available and kept for the purpose of extinguishing class A fires rapidly.
- Practice fire drills should be held so employees and boarders are familiar with their responsibilities should a real fire occur.

Lightning protection

- Buildings should be equipped with professionally installed lightning rods of copper or aluminum. The system should be properly grounded.
- All pipes, water systems, electrical systems and telephone lines should also be grounded.
- Contact a professional company for proper maintenance and installation.

When should you fight a fire?

In the event of a fire, your personal safety is your most important concern. You are not required to fight a fire. If all of the following conditions are met, then you may choose to use a fire extinguisher against the fire. If any of the conditions is not met, or you have even the slightest doubt about your personal safety, do not fight the fire.
Attempt to use a fire extinguisher if and only if...

- The fire alarm has been pulled and fire department has been called.
- The fire is small and contained.
- You know your escape route and can fight the fire with your back to the exit.
- You know what kind of extinguisher is required.
- The correct extinguisher is immediately at hand.
- You have been trained in how to use the extinguisher.

Electrical Safety

Electricity

Electricity is essential and considered as among the basic needs of everybody. Electricity had made our houses into homes, changed the mode of transportation from kalesas into taxis and Metro Railway Transport Systems (MRTS), and improved shops to malls and factories. It is hard to imagine if we had no electricity until now. However, it is also among the common causes of occupational accidents resulting to injuries, death and property damage.
More than a thousand workers are killed each year by electrical shock and thousands more are burned or maimed. More than 90% of the fatalities occurred when a person who was grounded made contact with live wire or an energized equipment housing. Line to line contact accounted for fewer than 10% of the deaths.

Electrical safety requires understanding of what electricity is, how electrical energy is transferred and how the path through which electrical current travels can be controlled.

Electric current requires a suitable circuit to provide the energy needed for lighting, heating, etc. An electrical circuit usually contains a power source and an electrical load. Suitable conducting material connects the power source to the load in order to complete the electrical circuit. These conductors are covered with a suitable insulating material to prevent the current from leaking out.

Some materials such as metals have loosely bonded electrons, and the amount of thermal energy available at room temperature is sufficient to generate free electrons. Materials that have a relatively large number of free electrons at room temperature, which are called **conductors**, are capable of conducting electricity (the movement of electrons in a given direction). On the other hand, materials that do not have a large number of free electrons at room temperature (such as plastics), which are called **insulators**, are incapable of conducting electricity. Materials that fall in between the two extremes are termed **semi-conductors**.

ELEMENTS OF ELECTRICITY

Voltage

In order for electrons to move between two points, a potential difference must exist. The potential difference between two points in a circuit is measured in terms of volts. The higher the potential difference, the easier it is for the electrons to move from one point to another, and the higher the electric current.

Resistance

The flow of electrons is also governed by the resistance offered by the conducting materials. It is measured in **Ohms**

Current

The current flow in a circuit is measure in terms of amperes. One ampere, by definition, is the flow of 6.28 x 1018 electrons per second past a given point in a circuit. Sometimes it is necessary to use smaller units of measurement for the current flow, the most commonly-used units being the milliampere (0.001 ampere)

Electrical resistance

Table 8.1 Normal Resistance Values of Various Materials

Material	Resistance (Ohms)
Most metals	>0 to 50
Dry wood	100,000
Wet wood	1,000
Dry Concrete on Grade	200,000 – 1,000,000
Wet Concrete on Grade	1,000 – 5,000
Leather Sole, dry, including foot	100,000 – 500,000
Leather Sole, damp, including foot	5,000 – 20,000

Table 8.2 Human Resistance to Electric Current

Body Area	Resistance (Ohms)
Human body, Internal (wet, ear to ear)	100
Human body , Internal (damp, hand to foot)	400 to 600
Human body (wet skin)	1,000
Human body (dry skin)	100,000 to 600,000

Source: Accident Prevention Manual

HAZARDS OF ELECTRICITY:

- Electric shock
- Burns
- Fire

Electric shock occurrence

Electrical shock is a common hazard encountered by people involved in the installation, maintenance, and operation of electrical equipment. Electric shock occurs once the worker' body becomes part of an electrical circuit when it comes in contact with a live internal conductor at the point of insulator breakdown.

The more common sources of electric shock are refrigerators and electric fans. Defective and poorly maintained electrical device will generate electrical leak. This leak passes all over the conductive materials of the device and if someone touches the device he will receive electric shock.

Below are the common causes of electrical injuries/accidents:
(a) touching of live parts
(b) short circuit
(c) inadequate guarding
(d) overloading
(e) breaking of connections

When the electric current has sufficient potential difference to overcome the body's resistance, it results in shock burns or even death. Although potential difference determines whether the body's resistances will be overcome, the damaging factor in electrical shock is the current flow.

FACTORS AFFECTING ELECTRIC SHOCK

1. Amount of current that flows through the human body. The amount of current that flows to the body depends on:
   - **Voltage of the circuit**. According to Ohms Law, voltage is directly proportional to the current. A higher voltage means a higher amount of current.
   - **Insulating quality**

2. The path the current takes through the body affects the degree of injury. A small current that passes from one hand to the other hand through the heart is capable of causing severe injury or death. However, there have been cases where larger currents caused an arm or leg to burn off without going through the vital organs of the body. In many such cases the person was not killed; had the same current passed through the vital organs of the body, the person easily could have been electrocuted.

3. Duration of current flow. The longer the current flows through the body, the more devastating the result can be. That is the reason why immediate action should be taken to free co-workers when they are shocked or burned by electricity.

ACTIONS TO TAKE

Shut off the electrical current if the victim is still in contact with the energized circuit. While you do this, have someone else call for help. If you cannot quickly get to the electrical disconnect to turn off the current, pry the victim from the circuit with something that does not conduct electricity such as a dry wood broom stick.

Do not touch the victim yourself if he or she is still in contact with an electrical circuit! You will become a victim of electrical shock.

4. Type of electric energy involved. There are two kinds of electrical energy:
   - **Alternating current (AC)** - the flow of electric charge whose magnitude and direction changes periodically. This can cause a person to maintain an involuntary grip on the live metal or conductor and prolong the current flow.
   - **Direct current (DC)** – the flow of electric charge that does not change direction

5. Body condition. Personal sensitivity to electric shock varies with age, sex, heart condition, etc. An electrical current passing through the body can cause severe injury or death by:
   - Contracting the chest muscles, resulting in breathing difficulty and death due to asphyxiation.
   - Affecting the central nervous system, resulting in malfunction of vital body function such as respiration
   - Interference with the normal rhythm of the heart beat, resulting in Ventricular Fibrillation which is defined as "very rapid uncoordinated contractions of the ventricles of the heart resulting in loss of synchronization between heartbeat and pulse beat." Once ventricular fibrillation occurs, it will continue and death will ensue within a few minutes.
   - Electricity may also affect the heart muscle, resulting in severe heart muscle contraction and cessation of heart action.
   - Heat generated when current overcomes tissue resistance may cause destruction of the body tissues.

   - The severity of an electric shock is the product of the current value and the time it flows through the body

   - **Let go current** – the maximum current that a person can tolerate when holding a conductor and can still free himself/herself by muscular stimulation.

   - **Ventricular fibrillation** – most death by electric shock are caused by ventricular fibrillation. It is a condition wherein the heart will not pulse regularly causing the heart to cease functioning. Once this occurs, the victim will be dead in a few minutes even if the electric source is interrupted.

   - Even small amounts of current can cause minor shock sensations and result to secondary accidents.

Based on the research of Charles F. Dalziel, professor at the University of California, the effects of alternating current (60Hz) on the human body are generally accepted to be as follows:

Current	Effect
0 – 1mA	No sensation, not felt
1 mA	Shock perceptible, reflex action to jump away. No direct danger from shock but sudden motion may cause accident.
> 3mA	Painful shock
6 mA	Let go current for women
9 mA	Let go current for men
>10mA	Local muscle contractions, sufficient to cause freezing to the circuit for 2.5% of the population
>15mA	Local muscle contractions, sufficient to cause "freezing" to the circuit for 50% of the population
>30mA	Breathing difficulty; can cause unconsciousness
50 – 100 mA	Possible ventricular fibrillation of the heart
100 – 200mA	Certain ventricular fibrillation of the heart
> 200mA	Severe burns and muscular contractions; heart more apt to stop than fibrillate
> 1A	Irreparable damage to body tissue

Source: Occupational and Environmental Safety Engineering and Management

There are four main types of injuries caused by electric currents **– electrocution (fatal), electric shock, burns**, and **falls.** These injuries can happen in various ways:

- direct contact with the electrical energy.
- when the electricity arcs (jumps) through a gas (such as air) to a person who is grounded (that would provide an alternative route to the ground for the electricity).
- thermal burns including flash burns from heat generated by an electric arc, and flame burns from materials that catch fire from heating or ignition by electric currents. High voltage contact burns can burn internal tissues while leaving only very small injuries on the outside of the skin.
- muscle contractions, or a startle reaction, can cause a person to fall from a ladder, scaffold or aerial bucket. The fall can cause serious injuries.

ELECTRIC SHOCK PREVENTION

(a) Use of grounding system

Grounding or earthing is any means of absorbing any leakage current and making it flow directly to earth by using an electrical conductor. It is a process of connecting metal parts/casing of the electrical equipment to earth through grounding wires. The voltage exists on the metal casing and earth resistance. Grounding means safety. There are two types of grounding:

(1) **System Grounding** – means grounding the neutral point iron terminal of electrical circuits on power transformer of electrical system;

(2) **Equipment Grounding** – grounding of a non-charged metal part of electrical equipment.

(b) Use Double Insulating Materials

Insulating materials have extremely high resistance values, virtually to prevent flow of electric current through it. The principle of insulation is used when work must to be carried out near un-insulated live parts. Work on un-insulated parts are carried out by using protective devices such as insulating stands, mats or screens, or rubber insulating gloves to protect workers from electric shock.

(c) Use Appropriate Disconnecting Means

(1) Fuse

A fuse is essentially a strip of metal that melts at a pre-determined value of current flow, and therefore cuts off the current to that circuit. In the event of abnormal conditions such as faults or when excess current flows, the fuse would blow and protect the circuit or apparatus from further damage. In effective and safe operation, the fuse should be placed in a live conductor and never in the neutral conductor. Otherwise, even with the fuse blown or removed, parts of the circuit such as switches or terminals will be affected. Over-fusing means using a fuse rating higher than that of the circuit it is meant to protect. This is dangerous because in the event of a fault, a current may flow to earth without blowing the fuse, endangering workers and the circuit or equipment concerned. It could also result in overheating of the cable carrying the excessive current, with the risk of fire.

(2) Circuit Breaker

A circuit breaker has several advantages for excess current circuit protection. The principle of the operation is that excess current flow is detected electromagnetically and the mechanism of the breaker automatically trips and cuts off electric supply to the circuit it protects.

(3) Earth Leakage Circuit Breaker

Majority of electric shock injuries occur when the body acts as conductor between line and earth. Protection against such shocks is provided by the inclusion of a current sensitive earth leakage circuit breaker (ELCB) in the supply line. ELCB may detect both over-current and earth leakage currents and thereby give very good circuit protection.

(d) Proper Maintenance of Portable Power Tools

The necessity to use flexible cables to supply electricity to the tools introduces hazards. Such cables are often misused and abused resulting in damaged insulation and broken or exposed conductors. The tool itself could also become charged with electricity due to a fault. Constant care and adequate maintenance and storage are essential to safe use.

CAUSES OF ELECTRICAL FIRE

The more frequent causes of electrical fires may be listed under three general classes namely, **arcs, sparks** and **overheating**. An **arc** is produced when an electric circuit carrying a current is interrupted, either intentionally – by a knife switch or accidentally – where a contact at a terminal becomes loose. The intensity of the arc depends, to a great extent, on the current and voltage of the circuit. The temperature of the electric arc is very high and any combustible materials in its vicinity may be ignited by the heat.

An electric arc may not only ignite combustible materials in its vicinity such as the insulating covering of the conductor, but it may also fuse the metal with the conductor. Hot sparks from burning combustible material and hot metal are thrown about, and may set fire to other combustible materials.

When an electric conductor carries a current, heat is generated in direct proportion to the resistance of the conductor and to the square of the current. The resistance of conductors is used to convey current to the location where it is used, or to convey it through the windings of a piece of apparatus, except in resistance devices and heaters.

HAZARDOUS LOCATIONS

Hazardous locations are areas where explosive or flammable gases or vapors, combustible dust, or ignitable fibers are present or likely to become present. Such materials can ignite as a result of electrical causes only if two conditions co-exist:

1. The proportion of the flammable substance to oxygen must permit ignition and the mixture must be present in a sufficient quantity to provide an ignitable atmosphere in the vicinity of electrical equipment.

2. An electric arc, flame escaping from an ignited substance in an enclosure, heat from an electric heater, or their source, must be present at a temperature equal to or greater than the ignition point of the flammable mixture.

Classification of Hazardous Locations

Class I – locations where flammable gases or vapors are present or likely to become present.

Class II – applies to combustible dusts.

Class III – locations are those where easily ignitable dust such as textile fibers are present but not likely to be suspended in the air in sufficient concentration to produce an easily ignitable atmosphere.

Explosion Proof Apparatus - A device enclosed in a case that is capable of:

- withstanding an explosion of a specified gas or vapor that may occur within it
- preventing the ignition of a specified gas or vapor outside the enclosure that may be caused by sparks, flashes or explosion of the gas or vapor inside the apparatus.

SAFE PRACTICES AND PROCEDURES

The following are the simple rules when working with electricity:

1. Always assume that a circuit is energized.
2. Use the appropriate instrument for testing circuits.
3. Use protective devices (ELCB, fuse, rubber mats, etc.).
4. Use personal protective equipment (rubber gloves, boots, safety devices).
5. Inspect tools, power cords, and electrical fittings for damage or wear prior to each use. Repair or replace damaged equipment immediately.

6. Use warning signs and isolate dangerous areas.
7. Observe proper maintenance schedules of electrical equipment, loads and wires.
8. Always tape cords to walls or floors when necessary. Nails and staples can damage cords causing fire and shock hazards.
9. Use cords or equipment rated for the level of amperage or wattage that you are using.
10. Always use the correct size fuse. Replacing a fuse with one of a larger size can cause excessive currents in the wiring and possibly start a fire.
11. Conduct regular electrical inspections for the electrical circuit.
12. Be aware that unusually warm or hot outlets may be a sign that unsafe wiring conditions exists. Unplug any cords to these outlets and do not use until a qualified electrician has checked the wiring.
13. Place halogen lights away from combustible materials such as cloths or curtains. Halogen lamps can become very hot and may be a fire hazard. Risk of electric shock is greater in areas that are wet or damp. Install Ground Fault Circuit Interrupters (GFCIs) as they will interrupt the electrical circuit before a current sufficient to cause death or serious injury occurs.
14. Make sure that exposed receptacle boxes are made of non-conductive materials.
15. Know where the breakers and boxes are located in case of an emergency.
16. Label all circuit breakers and fuse boxes clearly. Each switch should be positively identified as to what outlet or appliance it is for.
17. Do not use outlets or cords with exposed wiring.
18. Do not use power tools when protective guards are removed.
19. Do not block access to circuit breakers or fuse boxes.
20. Do not touch a person or electrical apparatus in the event of an electrical accident. Always disconnect the current first.
21. Ensure that only qualified personnel work on any part of an electrical circuit or equipment/apparatus.
22. Always replace a fuse with the same kind and rating. Never bridge a fuse using metal wires or nails, etc.
23. Make sure that there is someone to look after you whenever you work with any part of the electrical circuit.
24. Observe lock-out/tag-out (LOTO). Always lock safety switches and place tags before working on a circuit. Before energizing a circuit, ensure all personnel are clear of the circuit or the equipment concerned.
25. Ensure that temporary electrical installations do not create new hazards.
26. Always use ladders made of wood or other non-conductive materials when working with or near electricity or power lines.
27. Adhere to strictly established regulations of the Philippine Electrical Code.

TIPS FOR WORKING WITH POWER TOOLS:

- Switch tools OFF before connecting them to a power supply.
- Disconnect power supply before making adjustments.
- Ensure tools are properly grounded or double-insulated. The grounded tool must have an approved 3-wire cord with a 3-prong plug. This plug should be plugged in a properly grounded 3-pole outlet.
- Test all tools for effective grounding with a continuity tester or a ground fault circuit interrupter (GFCI) before use.
- Do not bypass the switch and operate the tools by connecting and disconnecting the power cord.
- Do not use electrical tools in wet conditions or damp locations unless tool is connected to a GFCI.
- Do not clean tools with flammable or toxic solvents.
- Do not operate tools in an area containing explosive vapors or gases.

TIPS FOR WORKING WITH POWER CORDS:

- Keep power cords clear of tools during use.
- Suspend power cords over aisles or work areas to eliminate stumbling or tripping hazards.
- Replace open front plugs with dead front plugs. Dead front plugs are sealed and present less danger of shock or short circuit.

- Do not use light duty power cords.
- Do not carry electrical tools by the power cord.
- Do not tie power cords in tight knots. Knots can cause short circuits and shocks. Loop the cords or use a twist lock plug.

A **Ground Fault Circuit Interrupter (GFCI)** - works by detecting any loss of electrical current in a circuit. When a loss is detected, the GFCI turns the electricity off before severe injuries or electrocution can occur. A painful shock may occur during the time that it takes for the GFCI to cut off the electricity so it is important to use the GFCI as an extra protective measure rather than a replacement for safe work practices.

SAMPLE CHECKLIST FOR BASIC ELECTRICAL SAFETY:

Inspect Cords and Plugs

- Check power cords and plugs daily. Discard if worn or damaged. Have any cord that feels more than comfortably warm checked by an electrician.

Eliminate Octopus Connections

- Do not plug several power cords into one outlet.
- Pull the plug, not the cord.
- Do not disconnect power supply by pulling or jerking the cord from the outlet. Pulling the cord causes wear and may cause a shock.

Never Break OFF the Third Prong on a Plug

- Replace broken 3-prong plugs and make sure the third prong is properly grounded.

Never Use Extension Cords as Permanent Wiring

- Use extension cords only to temporarily supply power to an area that does not have a power outlet.
- Keep power cords away from heat, water and oil. They can damage the insulation and cause a shock.
- Do not allow vehicles to pass over unprotected power cords. Cords should be put in conduit or protected by placing planks alongside them.

Commonly-Asked Questions and Answers

Q: How much electricity is enough to cause death?
A: It depends entirely on the amount of current flowing thru the body. At 10 milli-amperes, the body experiences involuntary grip or the person cannot let go of the conducting material. At 20 mA, the body may already experience ventricular fibrillation leading to death.

Q: Why do birds standing on high-tension wires do not get electrocuted?
A: In order for an animal or person be electrocuted, the electricity should have electric current path within the body or a closed circuit path for the electricity. The bird happens to be standing only on the wire without causing itself to create a circuit path.

Q: What can you say about the practice of replacing a busted fuse with any type of conductive material to continue the electrical supply?
A: The fuse is a protective device to prevent overload. Replacing it with other than the rated fuse with another type of conductor will defeat its purpose for protection. This practice will create the overload to the circuit that will cause damage to the electrical installation and in worst cases cause fire.

Q: What is the purpose of the third conductor of common plugs/third hole in other outlets? How come it is okay to only use the two holes or remove the third wire?
A: It is actually required for these kinds of electrical systems to have a ground wire or the third wire. This wire or conductor is for the ground electrical path. The ground electrical path is for the protection of the user or the

equipment itself. Removing the wire/conductor may cause damage to the equipment or cause electrical shock to the user.

Personal Protective Equipment

Defining hazards
What is a hazard? A hazard is anything that produces adverse effects on anyone. Examples of hazards are slippery floors, falling objects, chemicals and many more.

Classification of hazards
Hazards may be classified into direct, physical, chemical, biological and ergonomic.
A. **Direct hazards** – These are very common in companies that utilize oil, water or any liquid in the production process and in the construction industry where there are a lot of falling debris, like small pieces of wood, nails, and hand tools.
Examples:

- Unguarded moving parts of machines
- Falling/flying particles
- Slippery floors

B. **Physical hazards**
1. **Noise.**
The following table is the allowable time a worker can stay in a work area without hearing protection.
Allowable Exposure to Noise
8 hrs --- 90 db
4 hrs --- 95 db
2 hrs --- 100 db
1 hr --- 105 db
For an eight-hour exposure, the allowable noise level is 90 db.

2. **Extreme Temperatures** are of two types: **extreme heat** which can cause heat stroke and **extreme cold** which can cause hypothermia.

3. **Radiation** also has two types: the **ionizing radiation** and the **non-ionizing type**.
Ionizing radiation

- Ultraviolet (UV) light or alpha particle - from the sun can be shielded by paper
- Beta particle – can penetrate paper but not concrete. .
- Gamma ray – can penetrate concrete. This can be shielded by using lead like in the x-ray room which is made up of sheeted lead in-between concrete to prevent outside exposure.

Non-ionizing radiation

- radio waves, electric waves and infrared rays. An example is the welding process which produces infrared rays that can damage the skin.

Radiation is dangerous because it cannot be detected by the five senses but it destroys the cells and tissues of living organisms, and has long-term effects.
Three safety practices for controlling body exposure to radiation:

a. Time – the shorter the time, the lower the exposure received
b. Distance – the greater distance, the lower the exposure received
c. Shielding – may be lead, steel, iron or concrete

4. **Extreme Pressure** – These are pressures beyond the allowable levels needed by the human body. Normal atmospheric pressure is 14.7 psi, and even a small change in the atmospheric pressure has a corresponding effect to humans. Examples of workers exposed to extreme pressure are those involved in excavation work, scuba diving, and piloting airplanes.

5. **Vibration**

C. Chemical Hazards – These are substances in solid, liquid or gaseous forms known to cause poison, fire, explosion or ill effects to health. Examples include gases, fumes, vapor, mist and dust. These are airborne particles or airborne toxic elements that evaporate in the air and can cause irritation, discomfort and even death.
Chemical routes of entry to the body are by inhalation, ingestion and skin absorption.

D. **Biological Hazards** – These are hazards caused by viruses, fungi and bacteria.

E. **Ergonomic Hazards** – These are caused by improper posture or postural stress.

PPE – these are considered as the last line of defense. These devices provide limited protection to the ones using them.

Uses of PPE

Commonly used PPE in the workplace include: helmet, respirator, spectacles, earplugs, gloves, safety shoes, etc. The following are the functions and uses of PPE.

1. **Head Protection**

A safety hat is a device that provides head protection against impact from falling objects and protection against electrocution. Safety hats should be inspected prior to each use. Any one of the following defects is a cause for immediate removal of the PPE from service:

- Suspension systems that show evidence of material cracking, tearing, fraying or other signs of deterioration. Suspension should provide a minimum clearance of 1 to 1.25 in. (2.5 – 3.2 cm) between the top of the worker's head and the inside crown of the hat.
- Any cracks or perforations of brim or shell, deformation of shell, evidence of exposure to excessive heat, chemicals or radiation. Shells made of polymer plastics are susceptible to damage from ultraviolet light and gradual chemical degradation. This degradation first appears as a loss of surface gloss called chalking. With further deterioration, the surface will begin to flake away.

2. **Eye Protection**

A device that provides eye protection from hazards caused by:

- Flying particles
- Sparks
- Light radiation
- Splashes
- Gases

Goggles come in a number of different styles for a variety of uses such as protecting against dust and splashes: eye cups, flexible or cushioned goggles, plastic eye shield goggles and foundry men's goggles.

Eye protectors must meet the following minimum requirements:

- Provide adequate protection against the particular hazards for which they are designed
- Be reasonably comfortable when worn under the designated conditions
- Fit snugly without interfering with the movements or vision of the wearer
- Be durable
- Be capable of being disinfected
- Be easily cleaned
- Be kept clean and in good condition

3. **Face Shields**

Face shields should only be used as eye and face protection in areas where splashing or dusts, rather than impact resistance is the problem. In the case of grinding operations (plus other operations), a face shield is only secondary protection to other protective devices, such as safety goggles.

4. **Ear Protection**

Hazard:

- excessive noise - Noise exceeding 85-90 dB or more on eight hour exposure.

Examples:

- Ear plug
- Ear muffs
- Canal caps

The prevention of excessive noise exposure is the only way to avoid hearing damage. Engineering and administrative controls must be used if acceptable sound levels are exceeded. If such controls fail to reduce the sound levels to acceptable limits, personal hearing protection must be used.
Earmuffs must make a perfect seal around the ear to be effective.

5. **Respiratory Protection**

Respiratory protection is required when engineering improvements and administrative controls can't eliminate risk. Engineering controls include, isolation of the source of contaminants; design process or procedural changes, etc. Administrative controls on the other hand include, monitoring, limiting worker exposure, training and education, etc.

Hazards:

- Mists or Vapors
- Gases
- Smoke
- Fumes
- Particulates or dust
- Insufficient oxygen supply

Types of respirators are divided into two categories:

A. **Air purifying respirators**

- **Particulate respirators or mechanical filters** - screen out dust, fog, fume, mist spray or smoke. Such filters need to be replaced at frequent intervals.

- **Chemical cartridge devices** - remove contaminants by passing the tainted air through material that traps the harmful portions. There are specific cartridges for specific contaminants. These should be used and no substitutions should be made.

B. **Air supplying devices**

- Self-contained are those where the air supply is easily transportable and they protect against toxic gases and lack of oxygen. A common example is the self-contained breathing apparatus (SCBA), where the air tank is strapped to the wearer's back.
- Supplied-air respirators get air through an air line or hose. The breathable air is supplied by an air compressor or uncontaminated ambient air.

Air Contaminants – are divided into four types, gaseous, particulate, combination of gaseous and particulate and oxygen deficiency.

- **Gaseous contaminants** include gases and vapors.
- **Particulate contaminants** include dust, fumes, mist, fog and smoke.
- **Combination contaminants** usually consist of gaseous materials and particulates and result from operations such as paint spraying.
- **Oxygen-deficient atmospheres** are those that have less than 19.5 percent by volume. They often occur in confined spaces and are considered to be immediately dangerous to life and health.

6. **Hand and Arm Protection**

Hand and arm protection is required when workers' hands are exposed to hazards such as harmful substances that can be absorbed by the skin, severe cuts or lacerations, severe abrasions, chemical burns, thermal burns, and temperature extremes.

Examples of hand protection

- appropriate gloves
- hand pads
- barrier cream
- sleeves (for arm protection)

Hazards:

- Pinch points
- Hot surfaces
- Chemical substances
- Sharp objects
- Electrical

The following is a guide to the most common types of protective work gloves and the types of hazards they can guard against.

a. **Metal mesh, leather or canvas gloves** - Provide protection against cuts, burns, and sustained heat.

b**. Fabric and coated fabric gloves** - These gloves are made of cotton or other fabric to provide varying degrees of protection.

c. **Chemical and liquid-resistant gloves** - Gloves made of rubber (latex, nitrile, or butyl), plastic, or synthetic rubber-like materials such as neoprene protect workers from burns, irritation, and dermatitis caused by contact with oils, greases, solvents, and other chemicals. The use of rubber gloves also reduces the risk of exposure to blood and other potentially infectious substances.

7. **Foot and Leg Protection**

Hazards:

- Falling or rolling objects
- Sharp objects
- Hot surfaces
- Wet, slippery surfaces
- Electricity

Conductive Shoes protect against the buildup of static electricity or equalize the electrical potential between personnel and the ground. These shoes should be worn only for the specific task(s) for which they are designed, and should be removed at task completion and not used as general purpose footwear. This type of shoes must not be used by personnel working near exposed energized electrical circuits. Personnel must avoid wearing 100 percent silk, wool, or nylon hose of socks with conductive hose because these materials are static producers. Likewise, foot powders must be avoided because they are insulators and interfere with electrical conductivity.

Electrical Hazard Safety Shoes are non-conductive and protect against open circuits of 600 volts or less under dry conditions. The insulating qualities may be compromised if the shoes are wet, the rubber sole is worn out, or metal particles are embedded in the sole or heel. Electrical hazard shoes are not intended for use in explosive or hazardous locations where conductive footwear is required. This footwear should be used in conjunction with insulated surfaces.

8. **Fall Protection**

- **Travel restraint system** is an assembly composed of body belt and proper accessories that prevent a worker in a high elevation working area from traveling to an edge where the occurrence of fall may happen.
- **Fall arrest system** is an assembly composed of full-body harness, safety lanyard and proper accessories or a safety net which protect a worker after a fall by stopping the fall before hitting the surface below.
- **Lifelines** shall be secured above the point of operation to an anchorage or other structural member.

9. **Torso/ Full Body Protection** must be provided for employees if they are threatened with bodily injury of one kind or another while performing their jobs, and if engineering, work practices, and administrative controls have failed to eliminate these hazards.

Workplace hazards that could cause bodily injury include the following:

- Intense heat
- Splashes of hot metals and other hot liquids
- Impact from tools, machinery, and other materials
- Cuts
- Hazardous chemicals
- Contact with potentially infectious materials, like blood
- Radiation

As with all protective equipment, protective clothing is available to protect against specific hazards. Depending upon the hazards in the workplace, it may be needed to provide the workers with one or more of the following:

- Vest
- Jacket
- Apron
- Coverall
- Surgical gowns
- Full-body suits

These protective clothing come in a variety of materials, each suited to particular hazards. These materials include the following:

- **Paper-like fiber** - Disposable suits made of this material provide protection against dust and splashes.
- **Treated wool and cotton** - Adapts well to changing workplace temperatures and is comfortable as well as fire resistant.
- **Duck** - This closely woven fabric protects employees against cuts and bruises while they handle heavy, sharp, or rough materials.
- **Leather** - Leather protective clothing is often used against dry heat and flame.
- **Rubberized fabrics, neoprene, and plastics** - protective clothing made from these materials protect against certain acids and other chemicals.

Industrial Hygiene

Industrial hygiene is "the science and art devoted to the recognition, evaluation and control of environmental factors or stresses arising in or from the workplace, which may cause sickness, impaired health and well-being, or significant discomfort and inefficiency among workers or citizens of the community."

Implementing industrial hygiene practices such as exposure assessment and instituting control measures to minimize occupational accidents and diseases and their costs as well as enhance productivity. Industrial hygiene is interconnected with the different aspects of work – research and development, production, medical/health, safety and management.

RECOGNITION OF OCCUPATIONAL HEALTH HAZARDS

A. **Classification of occupational health hazards**

The various environmental stresses or hazards, otherwise known as occupational health hazards can be classified as chemical, physical, biological, or ergonomic.

1. **Chemical Hazards**. Occupational health hazards arise from inhaling chemical agents in the form of vapors, gases, dusts, fumes, and mists, or by skin contact with these materials. The degree of risk of handling a given substance depends on the magnitude and duration of exposure.

a. **Gases** are substances in gaseous state are airborne at room temperature. Examples are chlorine, hydrogen sulfide, ammonia, carbon monoxide, sulfur dioxide, phosgene and formaldehyde.
b. **Vapour** results when substances that are liquid at room temperature evaporate. Examples are the components of organic solvents such as benzene, toluene, acetone, and xylene.
c. **Mist** is a fine particles of a liquid float in air (particle size of 5 to 100 um approximately. Examples: nitric acid and sulfuric acid.
d. **Dust** is a solid harmful substances are ground, cut or crushed by mechanical actions and fine particles float in air (particle size of about 1 to 150 um). Examples are metal dusts and asbestos.
e. **Fume** is a gas (such as metal vapor) condensed in air, chemically changed and becomes fine solid particles which float in air (particles size of about 0.1 to 1 um). Examples are oxides generated from molten metal such as cadmium oxide, beryllium oxide, etc.

2. **Physical Hazards**. Problems relating to such things as extremes of temperature, heat stress, vibration, radiation, abnormal air pressure, illumination, noise, and vibration are physical stresses. It is important that the employer, supervisor, and those responsible for safety and health are on guard to these hazards due to the possible immediate or cumulative effects on the health of the employees.
a. **Extreme temperature**
Extreme temperatures (extreme heat and extreme cold) affect the amount of work that people can do and the manner in which they do it. In industry, the problem is more often high temperatures rather than low temperatures.
Heat stress may be experienced by workers exposed to excessive heat arising from work. Workers at risk of heat stress include outdoor workers and workers in hot environments such as firefighters, bakery workers, farmers, construction workers, miners, boiler room workers, factory workers, etc. workers aged 65 and older, those with heart disease, hypertension or those taking medications are at a greater risk for heat stress.
The factors influencing heat stress include:

- **Air Temperature** - known as the ambient room temperature.
- **Air Humidity** - the amount of water vapor or moisture content of the air.
- **Air Velocity** - the rate at which air moves and is important in heat exchange between the human body and the environment. because of its role in convective and evaporative heat transfer. Air movement cools the body by convection, the moving air removes the air film or the saturated air (which is formed very rapidly by evaporation of sweat) and replaces it with a fresh air layer, capable of accepting more moisture from the skin.
- Radiant Temperature - the thermal load of solar and infrared radiation in the human body.
- Clothing – working clothes style/design/mode and the type of fabric can affect the body heat temperature.
- Physical Workload - may be categorized as light, moderate, or heavy depending on the task or job activity carried out by the worker.
  Light - work-sitting or standing to control machines.
  Moderate work - walking about, moderate lifting and pushing
  Heavy work – intense work of the extremities and trunk.

Cold stress. Workers exposed to extreme cold or work in cold environments such as those in ice plants or refrigerated workplaces may be at risk of cold stress.

Refrigerants such as ammonia, methyl chloride and halogenated hydrocarbons used in freezing and cold storage bring risks of poisoning and chemical burns. Ammonia and other refrigerants such as propane, butane, ethane and ethylene, though less frequently used are flammable and explosive chemicals. Monitoring and evaluation of these chemicals is highly recommended when working in cold storage and refrigerating plants.

b. **Radiation**
Types of Non-Ionizing Radiation
1. Ultraviolet (UV) Radiation
2. Infrared (IR) Radiation
3. Laser Radiation
4. Microwave Radiation

Effects of Non ionizing radiation
Ozone may be produced as a result of electrical discharges or ionization of the air surrounding non-ionizing radiation sources, e.g. UV, high power laser, microwave and short duration exposure in excess of a few tenths ppm can result in discomfort (headache, dryness of mucuous membranes and throat).

c. **Extreme pressure**
It has been recognized as from the beginning of caisson work (work performed in a watertight structure) that men working under pressures greater than at a normal atmospheric one, are subject to various illnesses connected with the job. Hyperbaric (greater than normal pressures) environments are also encountered by divers operating under water, whether by holding the breath while diving, breathing from a self-contained underwater breathing apparatus (SCUBA), or by breathing gas mixtures supplied by compression from the surface.
Occupational exposures occur in caisson or tunneling operations, where a compressed gas environment is used to exclude water or mud and to provide support for structures. Man can withstand large pressures due to the free access of air to the lungs, sinuses, and middle ear.

d. **Inadequate illumination**
The measure of the stream of light falling on a surface is known as illumination. The key aspects of illumination include lux, luminance, reflectance, glare and sources of lighting.

KEY ASPECTS OF ILLUMINATION:

- **Lux** - unit of measurement.
- Luminance - measure of light coming from a source
- Reflectance - ability of a surface to return light.
- Glare is caused by bright light sources which can be seen by looking in the range from straight-ahead to 450 above the horizontal. There are two types of glare: direct and reflected.
  Direct Glare is produced when light is positioned at the surface. It can be prevented by correct installation of lighting fittings, installing louvers below the light source, enclosing the lamps in bowl reflectors, and opaque or prismatic shades.

  Reflected Glare is produced when light is reflected off a shiny surface. It may be addressed by providing indirect lighting.

Sources of light

There are two sources of light:

- Daylight, also called natural light depends on the availability at the location and weather condition.
- Electric Light can come from:
  Incandescent lamps or bulbs
  Fluorescent lamps or tubes
  High intensity discharge or mercury

Types of Lighting

Illumination can also be viewed in terms of:

- General lighting illuminating the entire premises
- Local lighting directing light on a particular object that you are working with.

Factors in determining the quantity of light:

- Nature of work - more light will be required if one is working with small objects.
- Environment - the ability of the surrounding surfaces to reflect light.
- Eyesight of the workers - the ability of the eye to adjust rapidly to different distances declines as people grow older.

e. **Excessive vibration**

A body is said to vibrate when it is in an oscillating motion about a reference point. The number of times a complete motion cycle takes place during the period of one second is called the frequency and is measured in hertz (Hz). Vibration usually refers to the inaudible acoustic phenomena, which are recognized by through touch and feeling. It is a vector quantity described by both a magnitude and direction.

Portable meters are available for vibration measurements. These usually provide readouts that must be compared to the appropriate standards.

f. **Noise or unwanted sound** is a form of vibration conducted through solids, liquids, or gases. The level of noise in an industrial operation can constitute a physical hazard to the exposed workers. The extent of the hazard depends not only on the overall noise level but also on the time period and frequency and type of noise to which the worker is exposed.

Types of noise

- Continuous noise is a steady state noise with negligible level fluctuations during the period of observation.
- Intermittent noise levels shift significantly during observation.
- Impact noise consists of one or more bursts of sound energy, each lasting less than one second.

Factors that can influence noise exposure

A number of factors can influence the effects of the noise exposure.

These include:

- variation in individual susceptibility
- the total energy of the sound

- the frequency distribution of the sound
- other characteristics of the noise exposure, such as whether it is continuous, intermittent, or made up of a series of impacts
- the total daily duration of exposure

3. **Biological Hazards**
Biological hazards are any virus, bacteria, fungus, parasite, or living organism that can cause a disease in human beings. They can be a part of the total environment or associated with certain occupations such as medical professions, food preparation and handling, livestock raising, etc.

Diseases transmitted from animals to humans are commonly infectious and parasitic which can also result from exposure to contaminated water, insects, or infected people.

4. **Ergonomic Hazards**
"Ergonomics" literally means the customs, habits, and laws of work.
The ergonomics approach goes beyond productivity, health, and safety. It includes consideration of the total physiological and psychological demands of the job upon the worker. It deals with the interaction between humans and traditional environmental elements as atmospheric contaminants, heat, light, sound, and all tools and equipment used in the workplace.

Examples of ergonomic hazards are:
- Poor workplace design – cramped leg area, crowded worktable, distant work materials
- Awkward body postures – prolonged sitting, twisted body while bending
- Repetitive movements – sewing, cutting, stamping
- Static posture – prolonged standing without motion
- Forceful motion – extreme pulling and pushing
- Manual handling – improper carrying of materials, use of pliers.

In a broad sense, the benefits that can be expected from designing work systems to minimize ergonomic stress on workers are as follows:
- more efficient operation;
- fewer accidents;
- lower cost of operation;
- reduced training time; and
- more effective use of personnel

5. **Special Considerations:**
The following items have become important OSH issues that need to be addressed by Industrial Hygiene professionals and employers:
a. **Confined space** is an enclosed or a partially enclosed space. It has restricted entrance and exit (by location, size, and means) thus, the natural airflow is limited. This can cause accumulation of "dead" or "bad" air" and airborne contaminants. Confined spaces are not designed, and intended for human occupancy.

Examples of confined space:

Storage tanks, sewers, boilers, manholes, tunnels, pipelines, trenches, pits, silos, vats, utility vaults, culverts.

Hazards of confined space:

- **Oxygen deficiency** – air is considered oxygen deficient when the oxygen content is less than 19.5% by volume. Oxygen level in a confined space can decrease due to consumption or displacement by inert gases such as carbon dioxide or nitrogen. Work processes such as welding, cutting or brazing, and certain chemical reactions such as rusting and bacterial reaction (fermentation) can also reduce oxygen concentration.

- **Flammable/explosive atmosphere** – may result from:
  - Oxygen enriched atmosphere exists where oxygen in the air is greater than 21%. An oxygen enriched atmosphere will cause flammable materials such as clothing and hair to burn violently when ignited.
  - Flammable gas, vapor, or dust in proper proportion.

- **Toxic atmospheres** are those which contain toxic substances in concentrations that exceed the Threshold limit Value 9TLV), as specified in the Occupational Safety and Health Standards or the Chemical/Material Safety Data Sheet of the substance used at work.

  Toxic substances in the atmosphere may come from the following:
  - Products stored in the confined space
  - Work being performed in a confined space
  - Areas adjacent to the confined space

- **Mechanical and physical hazards**
  - Rotating or moving mechanical parts or energy sources can create hazards within a confined space
  - Physical factors such as extreme temperatures, noise, vibration and fatigue
  - Loose materials such as fine coal, sawdust or grains can engulf or suffocate the workers

b. **Indoor Air Quality (IAQ)** refers to the quality of the air inside buildings as based on the concentration of pollutants & thermal (temperature & relative humidity) conditions that affect the health, comfort and performance of occupants.

Sources of IAQ problems:

- Ventilation system deficiencies
- Overcrowding
- Tobacco smoke
- Microbiological contamination
- Outside air pollutants
- Off gassing from materials in the office, furniture and mechanical equipment.
- Poor housekeeping

Indoor air pollutants:

- Volatile Organic Compounds (VOCs)

- Formaldehyde
- Carbon Dioxide
- Carbon Monoxide
- Nitrogen Oxides
- Sulfur Dioxide
- Ammonia
- Hydrogen Sulfide
- Dust

Other factors affecting occupants:
- Comfort problems due to improper temperature and relative humidity conditions
- Poor lighting
- Unacceptable noise levels
- Adverse ergonomic conditions and
- Job-related psycho-social stressors.

OCCUPATIONAL HEALTH

Definition of Occupational Health
Occupational health has been defined by the ILO and the WHO as the
- Promotion and maintenance of the highest degree of physical, mental & social well-being of workers of all occupations
- Prevention among its workers of departures from health caused by their working conditions
- Protection of workers in their employment from risks usually from factors adverse to health
- Placing & maintenance of the worker in an occupational environment adapted to his/her physiological ability

As you know,
- Occupational health encompasses the social, mental and physical well-being of workers in all occupations.
- It includes the protection of workers from illnesses arising from work through promotion of safety and health programs.
- It should always be a priority to adapt the work to the human being. Poor working conditions have the potential to affect a worker's health and safety.
- Poor working conditions can affect not only the workers but their families, other people in the community, and the physical environment.

Key Concepts in Occupational Health
Remember that workplace hazards can potentially cause harm to a worker. However, the risk or the likelihood that this harmful effect would take place depends on the conditions of exposure. These factors include intensity and duration of exposure to the hazards, timing of exposure and multiplicity of exposure.
1. **Exposure duration or the length of time of being vulnerable to work hazards.**

Constant exposure to amounts which have low levels in the workplace over a prolonged period of time increases the risk of disease after a latency period (the interval between exposure to a hazard(s) and the clinical appearance of disease);

2. **Magnitude, level or dose of exposure.**
As the concentration or amount of a hazard is increased the likely it can do more harm.

3. **Timing of exposure**. This is related to exposure duration. A worker who is exposed to a hazard continuously or for several periods in a day is more at risk than those with less exposure

4. **Multiplicity of exposure**. Exposures to mixtures of hazards or several chemicals at the same time can cause synergistic or cumulative effects.

HEALTH EFFECTS OF BIOLOGICAL HAZARDS
Biologic hazards are plants, animals and their products that may present risks to the health of persons infected by biologic agents they carry. Such biologic agents are classified as bacteria, virus, fungi, and parasites depending on their physical and other cellular characteristics. For example, bacteria and fungi have cell walls while viruses do not.

TUBERCULOSIS
Tuberculosis is at the top of the list because it remains one of the most prevalent illness affecting Filipinos. It is among the leading causes of morbidity and mortality based on Philippine Health Statistics and Field Health Service Information System.

Tuberculosis (TB) is the sixth leading cause of illnesses and deaths in the Philippines; the country is ninth out of the 22 highest TB-burden countries in the world and has one of the highest burdens of multidrug-resistant TB.

Tuberculosis is a long-standing infection caused by the bacteria Mycobacterium tuberculosis. The TB bacteria usually attack the lungs, but can also attack any part of the body such as the kidney, spine, brain, bones and intestines. If not treated properly, TB disease can be fatal.

TB is primarily an airborne disease. The bacteria are spread from person to person in tiny microscopic droplets or aerosol when a TB sufferer forces air from his/her lungs when coughing, sneezing, speaking, singing, or laughing. A person then inhales the bacteria vigorously expelled from the lungs of an active TB patient. A person needs only to inhale a small number of these to be infected.

TB can survive for extended periods of time in the air and on various surface areas. It was found that 28% of the tuberculosis bacteria remain alive in a room after nine hours. Tuberculosis can live up to 45 days on clothing, 70 days in carpet, 90 to 120 days in dust, approximately 105 days on a paper book, and approximately six to eight months in sputum. Ultraviolet light, volume of air in a room and recirculation of air through a HEPA filter are important factors that affect the survival of the bacteria. Until the droplet falls, it can be breathed in at any time.

Symptoms do not appear unless a patient has active TB. The most common symptom of active pulmonary tuberculosis is coughing that lasts two or more weeks. Other symptoms are low grade fever, night sweats, feeling weak and tired, losing weight without trying, decreased or no appetite, chest pains and coughing up blood.

Only people with active TB whose sputum is contains the TB bacilli can spread the disease to others. However, exposure does not necessarily mean one will become infected with tuberculosis. If you have been exposed to an active tuberculosis patient or an area that is contaminated, you should seek medical advice immediately.

It is therefore best to understand the life cycle of tuberculosis. An index case or the first patient in an outbreak may spread the infection to people they spend time with every day. This includes family members, friends, and coworkers. However, people infected with the TB bacilli will not necessarily become sick with the disease. The immune system "walls off" the TB bacilli which, protected by a thick waxy coat, can lie dormant for years. About 90-95% undergo healing of their initial infection and the bacteria eventually die off. This stage manifests no symptom and is not contagious. This condition is known as inactive or latent TB. If, however, the body's resistance is low because of aging, infections such as HIV, malnutrition, or other reasons, the bacteria may break out of hiding and cause active TB in 5-10% of patients. Active TB patients manifest symptoms and become sick during their life while 30% of them, even if left untreated, will spontaneously remit or restore back to being healthy.

A person with TB disease, if left untreated, could infect approximately 10-20 persons within two years' time. The most infectious are the sputum smear (+). Persons with contagious TB disease must be treated and cured to stop the spread of TB in our communities. Active TB can also spread to other parts of the body through the bloodstream.

The best way to prevent tuberculosis is to strengthen one’s immune system by eating healthy, exercising regularly and getting plenty of rest. If one has active TB, covering the mouth when coughing, sneezing or laughing is one way to help prevent the propulsion of the bacteria into the environment.

Wearing a mask during this time is very helpful as well as staying at home until one’s sputum examination has reverted back to normal as certified by one’s physician. At home, sleep in a room by oneself to help prevent the transmission of the disease to other members of the family. Making sure that the workplace has proper ventilation is one way of preventing the transmission of the disease to co-workers.

HIV and AIDS

Another biologic hazard that has an impact in the workplace is HIV and AIDS though workers in the manufacturing sector are hardly exposed to this virus by the nature of their work. The Human Immunodeficiency Virus (HIV) is the cause of Acquired Immune Deficiency Syndrome (AIDS) - a condition in which progressive failure of the immune system allows life-threatening opportunistic infections and cancers to thrive.

HIV belongs to a group of viruses known as retroviruses. Once inside the human body, the virus attacks the cells of the immune system, specifically the CD4+ helper cells. The virus uses the helper cells’ cellular material or genetic material to replicate itself consequently filling the helper cell. The body tries to keep up by making new cells or trying to contain the virus, but eventually the HIV wins out and progressively destroys the body's ability to fight infections and certain cancers. HIV is present blood, semen, vaginal fluids and breast milk of an infected mother in infectious quantity.

AIDS is the result of HIV’s attacks on the body’s immune system. This medical condition leaves the individual so unprotected that any other virus that attacks the body can cause grave damage. It is the most advanced stage of infection with HIV.

Contrary to some popular misconceptions, HIV is a difficult disease to get. For HIV to be transmitted, the following three conditions must be met:

1. **HIV must be Present**

Infection can only happen if one of the persons involved is infected with HIV. Some people assume that certain behaviors (such as anal sex) cause AIDS, even if HIV is not present. This is not true.

2. **There must be a sufficient quantity of HIV**

The concentration of HIV determines whether infection may happen. In blood, for example, the virus is very concentrated. A small amount of blood is enough to infect someone. A much larger amount of other fluids would be needed for HIV transmission.

Blood contains the highest concentration of the virus, followed by semen, then by vaginal fluids. Breast milk can also contain a high concentration of the virus, but in this situation, transmissibility depends on WHO and HOW. An adult can ingest a small amount of breast milk at no probable risk. But an infant, with his/her very small body and newly forming immune system, consumes vast quantities of breast milk relative to his/her body weight. Therefore an infant is at risk from breast milk, whereas an adult may not be.

3. **HIV must get into the bloodstream**

It is not enough to be in contact with an infected fluid to become infected. Healthy, unbroken skin does not allow HIV to get into the body; it is an excellent barrier to HIV infection. HIV can only enter the bloodstream through an open cut or sore, or through contact with the mucous membranes or damaged tissue in the anus and rectum, the genitals, the mouth, and the eyes; or be directly injected from a needle or syringe.

You cannot get HIV from kissing, hugging, casual contact, drinking from the same glass, eating together, swimming pools, public toilets, pets, and mosquito bites. HIV is not transmitted through casual, every day contact. Since HIV is not transmitted by saliva, it is impossible to get it through sharing a glass, a fork, a sandwich, or fruit. The chemicals used in swimming pools and hot tubs would instantly kill any HIV, if the hot water had not killed it already.

Sterilized needles are always used in taking blood from donors, so HIV is not spread in this manner. Humans are the only animals that can carry HIV. HIV is not transmitted by mosquitoes, flies, ticks, fleas, bees or wasps. If a bloodsucking insect bites someone with HIV, the virus dies almost instantly in the insect's stomach (as it digests the blood). HIV can only live in human cells.

HIV antibody tests can have two different results: positive or negative.

1. A **positive result** on a confirmed HIV antibody test means that HIV antibodies are present and one is infected with HIV (called "HIV positive"); the person can infect others but does not mean that the person has AIDS.

2. **A negative result** on an HIV antibody test means that most likely one is not infected with HIV. However, it can take 3 to 6 weeks, and sometimes up to 3 months (and in few cases up to 6 months) before HIV

antibodies show up on a standard test. As a result, some people who are recently infected with HIV may still have a negative test result during this time.

Knowing your status can allow you to begin treatment which can help prevent the further spread of the virus, and in some cases prevent complications associated with HIV infection. HIV testing should be voluntary and confidential. Counseling before and after HIV testing will help you understand what behaviors put you at risk and teach you how to decrease the chance of becoming infected. If the test result is positive, counseling will address your immediate needs for support and information, and teach you on how to decrease the chance of infecting others.

HIV and AIDS is an urgent issue for the workplace that we must pay attention to because it has the potential to reduce productivity and economic growth. The virus can place heavy financial and social burden on families as they are faced with reduced income and often need to pay for an array of medical treatments. Workers and families also face considerable stigma and discrimination from the virus causing loss of jobs and other acts of discrimination in the community.

The **ABC approach** to preventing the sexual transmission of HIV has been defined and adopted by a variety of organizations, governments and non-governmental organizations ever since the term was first used in 1992 when the then Secretary of Health, seeking a compromise between the Catholic Church and government at the time brought together abstinence, fidelity and condom use to create the 'ABC slogan'.

According to UNAID's 2004 Global Report on the AIDS Epidemic, 'ABC' stands for:

- **Abstinence (not engaging in sex, or delaying first sex)**
- **Being safer, by being faithful to one's partner or reducing the number of sexual partners**
- **Correct and consistent use of condoms**

These were later expanded to include:

- **Don't share needles/sterilized needles**
- **Education and information**

TETANUS

Tetanus, commonly called lockjaw, is another illness caused by a bacterial toxin or poison from the spore of the bacterium Clostridium tetani.

Tetanus affects the nervous system and is characterized by an increase in muscle tone causing painful spasms. Lockjaw is one of the usual manifestations of untreated tetanus infection. Severe spasms and convulsions may occur which can lead to death if not treated early.

Workers engaged in agriculture have an increased risk of contracting tetanus. Such workers are particularly susceptible to cuts and abrasions, and owing to the nature of the work environment it is likely that wounds will become contaminated with soil containing tetanus spores. Other susceptible workers include miners and construction workers.

The illness is contracted through a cut or wound that becomes contaminated with tetanus bacteria. The bacteria can get in through even a tiny pinprick or scratch, but deep puncture wounds or cuts like those made by nails or knives are especially susceptible to infection. Tetanus bacteria are present worldwide and are commonly found in soil, dust and manure. Tetanus causes severe muscle spasms, including “locking” of the jaw so the patient cannot open his/her mouth or swallow, and may lead to death by suffocation.

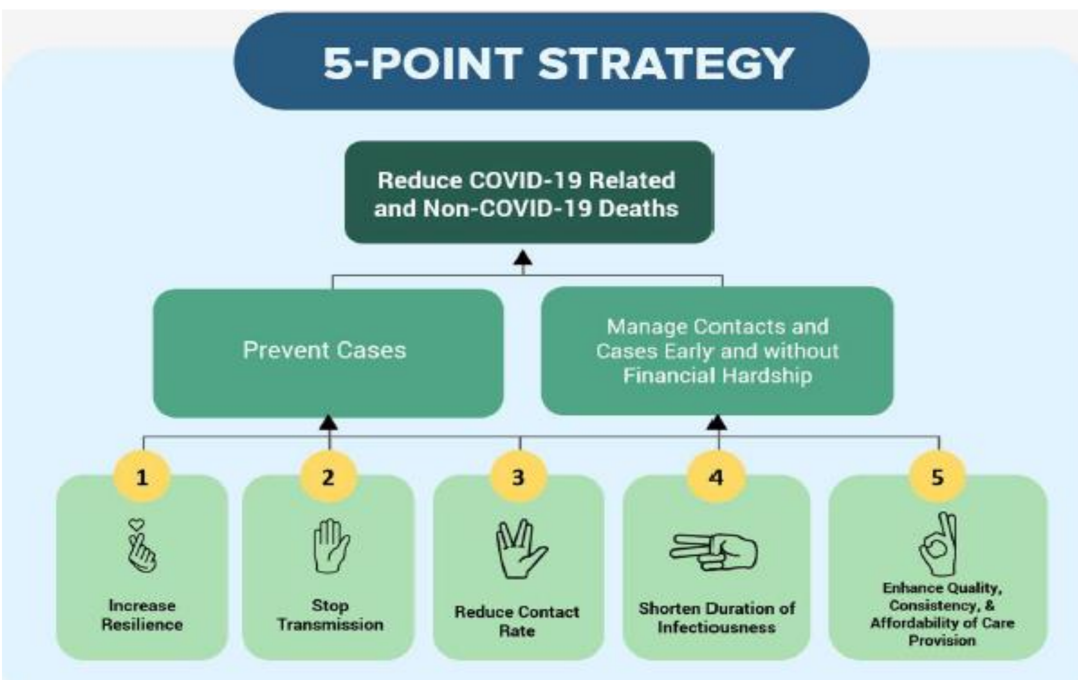
Common first signs of tetanus include muscular stiffness in the jaw (lockjaw) followed by stiffness of the neck, difficulty in swallowing, rigidity of abdominal muscles, generalized spasms, sweating and fever. Symptoms usually begin 7 days after bacteria enter the body through a wound, but this incubation period may range from 3 days to 3 weeks.

Tetanus is not transmitted from person to person. Effective prevention of tetanus is achieved by active immunization, using tetanus toxoid. In view of the substantial risk of contracting tetanus following an injury and in view of the fatal consequences of the disease, mass immunization is fully justified. Immunization is usually carried out as part of the occupational health program for workers. Active immunity is preferable to passive immunity, because there are fewer side reactions and because better protection is afforded.

A GUIDE TO THE NEW NORMAL

WHAT IS THE NEW NORMAL?

The New Normal refers to the emerging behaviors, situations, and minimum public health standards that will be institutionalized in common or routine practices and remain even after the pandemic while the disease is not totally eradicated through means such as widespread immunization.



The DOH's 5-point strategy is our plan of action in establishing the new normal in our daily routines.
The new normal revolves around the four strategies of the DOH's 5-point strategy to reduce COVID 19 related and non COVID 19 deaths - (1) increase resilience, (2) stop transmission, (3) reduce contact, (4)

shorten duration of infectiousness. With this, the DOH has developed minimum public health standards across different settings to aid the general public and the different sectors as we transition into the new normal.

RESPIRATORY HYGIENE AND COUGH ETIQUETTE

- Respiratory hygiene and cough etiquette refer to the set of practices that help prevent and control the spread of respiratory illnesses. This can range from covering your mouth when coughing to using tissue when sneezing
- The COVID-19 virus is transmitted through respiratory droplets. Respiratory droplets are small, wet droplets that come from our nose, mouth, throat and lungs. These droplets may be so small that they may not be seen by the naked eye. In light of the emerging evidence, it is suggested that airborne transmission of the virus may also occur. Practicing respiratory hygiene and cough etiquette may help limit the spread.
- You should cover your mouth and nose with a disposable tissue or the inner portion of the elbow when you sneeze or cough. Turn away from the people surrounding you and, if possible, distance yourself when you get the urge to sneeze or cough. After coughing or sneezing, dispose of the tissue that you used and wash your hands with soap and water. Use an alcohol-based hand sanitizer if soap and water is unavailable.

REDUCE EXPOSURE OF VULNERABLE INDIVIDUALS

- You are a vulnerable individual if you are pregnant, above 60 years old and/or have underlying health conditions such as cardiovascular disease, chronic respiratory disease, diabetes, hypertension and others
- Vulnerable individuals are more at risk of contracting severe COVID-19 than others. 10 This is because their condition makes them more susceptible to complications that can arise from the disease.
- If you are pregnant, above 60 years old or if you have any underlying medical conditions, stay home and limit your travel to essential services and urgent needs. lfyou don't have any of the conditions mentioned above, help those who do by going to the grocery or doing any other essential needs in their place

HANDWASHING

- Hygiene can be best described as the set of practices that help people prevent disease and stay healthy. Handwashing is an example of a practice of personal hygiene.
- The COVID-19 virus is spread through respiratory droplets. These droplets can linger on our skin and on surfaces. We may be able to be infected when contaminated hands touch our nose, mouth, or eyes or we may further spread the virus when we touch different objects or surfaces.

ENVIRONMENTAL HYGIENE

- Apart from our personal hygiene, we must keep our environment clean. Maintaining a healthy environment plays a key role in ensuring people are healthy and free from disease. Environmental surfaces include electronic devices, furniture, toilets and sinks, etc

- Respiratory droplets that land on surfaces may still remain infectious even after several hours and can last up to several days. With this, cleaning and disinfecting surfaces, especially frequently touched surfaces, may help mitigate the spread of the virus

- Clean surfaces with water and any household soap or detergent. Afterward, disinfect with chemical disinfectants such as household bleach.

To make your own household disinfectant solution, you may:
1. Using commercially available household bleach at 5% active chlorine, dilute 1 part of bleach to 9 parts clean water; or
2. Using chlorine powder I granules I tablets 60%-70% active chlorine,dissolve tablespoons of chlorine(equivalent to 10 grams) to liters of clean water. Mix the solution thoroughly using a stick

PROPER HANDWASHING

ON THE USE OF FACE MASKS

All individuals must wear face mask or appropriate Personal Protective Equipment (PPE) outside of residence, except in the following situations:

WHY?

1. Individuals eating or drinking provided there is at least two (2) meter distancing between persons and in a designated dining area;

2. Individuals engaging in high intensity physical activities whereby only those who are actively exercising may be allowed not to wear masks during the physical activity provided that at least two (2) meters physical distancing from other people is observed; and
3. Children under the age of two (2) years old, persons who are unconscious, incapacitated, or otherwise unable to remove the mask are not recommended to wear masks due to risks of suffocation

The droplets that we expel may be contained in our face masks and thus prevent contaminating our surrounding s. As infected, asymptomatic individuals may also transmit the virus, we do not know who may have COVID-19, it is encouraged that everyone wear face masks.

HOW TO WEAR YOUR MASK

1. Always wash your hands properly prior to wearing your mask and prior to removing it. If handwashing stations unavailable, clean your hands using alcohol based sanitizer prior to wearing your mask
2. Always make sure your mask is clean the mask prior to using it.
3. When putting on or removing your mask, you should always hold by the straps to avoid contamination.
4. The mask should be able to cover your mouth, nose and chin adjusting it accordingly to fit your face ensuring no air escapes the sides of the mask.
5. While removing the mask, always pull it away from your face to prevent contamination.
6. Wash/ sanitize your hands after removing your mask.

GUIDING PRINCIPLES

1. Masks should only be used by one person and should not be shared.
2. Masks should be replaced if wet or visibly soiled.
3. Masks should not be used for extended periods of time, if possible.
4. Damaged masks or masks which do not appropriately fit the user should not be used.
5. Never leave your mask on any surface when not in use.
6. Avoid touching the mask while wearing it.
7. Do not remove your mask in public spaces.
8. Reusable cloth masks should be washed at least once a day, ideally with hot water and soap/detergent.

HOW TO DISPOSE OF YOUR MASK

1. If and when possible, washing the cloth masks prior to disposal may lessen risk of contamination.
2. All masks shall be tightly sealed in plastic bags prior to disposing them in disposal units.
3. Individuals confirmed to have COVID-19 should separate their masks disposal into sealed yellow medical grade trash bags for disposal.
4. After disposing of any face mask, always wash your hands properly.

PRACTICE PHYSICAL DISTANCING

- Physical distancing means maintaining at least a one (1) meter space from other individuals and avoiding large gatherings of people as much as possible.
- Higher risk of COVID-19 transmission occurs between people who are within 1 meter of each other.
- Always plan ahead when going out for essential needs. Before heading out, think about the modes of transportation, methods for obtaining essential items, nature of social activities, etc. that allow for physical distancing. Avoid crowded areas and stay at least one (1) meter from other people. Whenever possible, stay at home.

RATIONAL USE OF FACE MASKS

Target Personnel/ Population	Setting	Type of face mask
General Population, above the age of 2 years	Public places, High density communities, Offices/ Workplaces, Food and Other Service Establishments, Schools, Hotels and other accommodations, Transportation and Ports of Entry, Churches/ Places of Worship, Prison/ places of detention	Non-medical face mask (i.e. cloth mask)
Patients with respiratory symptoms	Any setting	Medical-grade face mask (i.e N88 mask, surgical mask)
Caregiver of patients with respiratory symptoms	Any setting	
Healthcare workers and other frontliners	Health care facilities	Medical-grade face mask N95 mask (If with aerosolizing procedures)

DETECTION AND ISOLATION OF SYMPTOMATIC INDIVIDUALS

Feeling sick? You probably have to ask yourself if you've got:

- Cough or colds
- Sore throat
- Fever or chills
- Loss of smell or taste
- Body aches

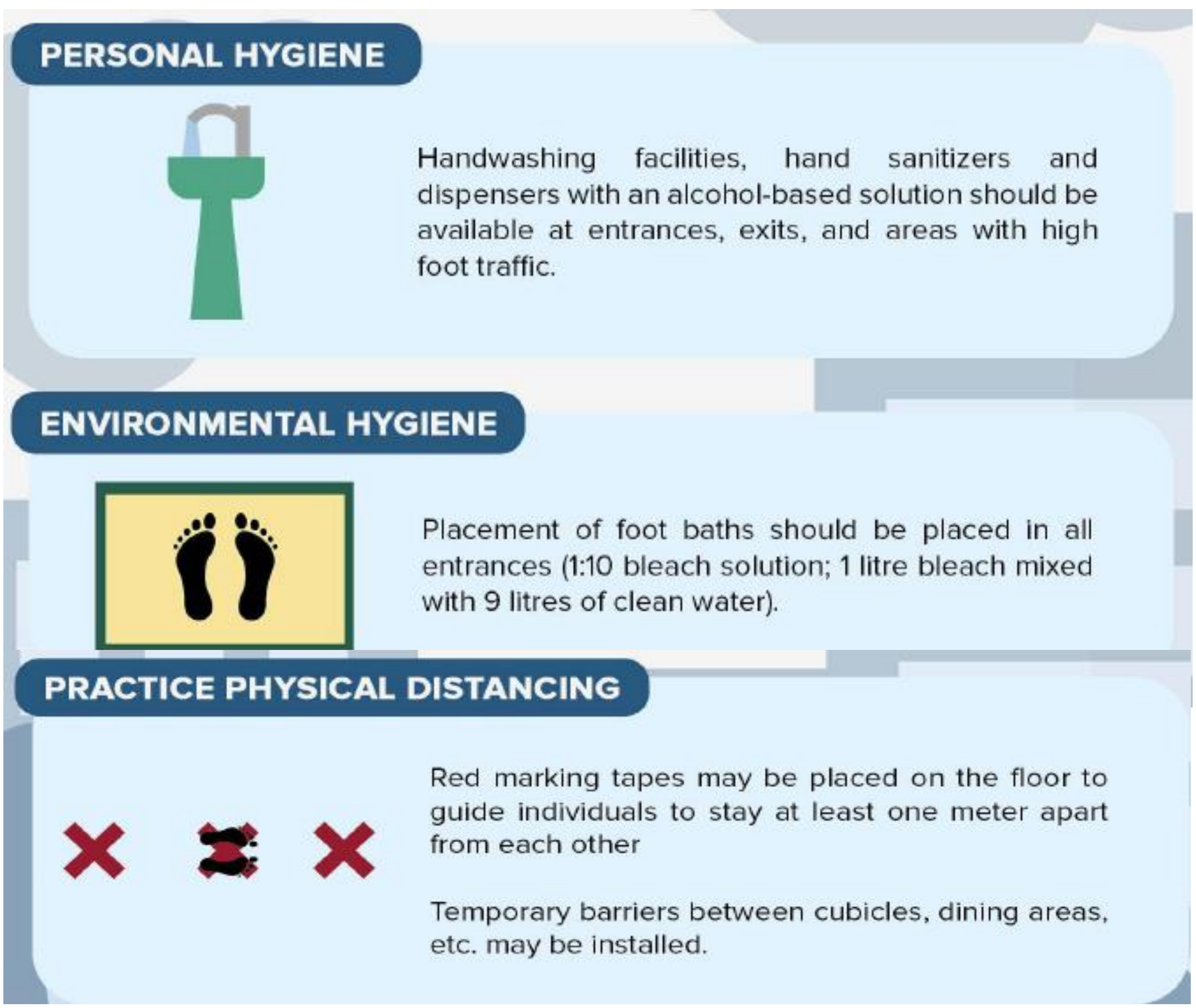
- Nausea or vomiting

If you've said yes to all or any of the above, then it's best to stay at home and report or call the medical officers in your barangay or municipality.

➢ Preventing COVID-19 transmission is best achieved by identifying suspect cases and properly referring them to appropriate primary care facilities immediately for consultation and assessment. Timely testing and isolation of suspected cases is critical to stop the transmissions of disease in the community

1. Always do a self-assessment of your symptoms
2. Keep yourself updated on the latest developments of COVID-19.
3. Know the nearest health facility you can go to if your symptoms get worse.
4. Be honest when consulting with health professionals, they can help you more this way.
5. If you are COVID-19 positive, isolate in a quarantine facility or at home* for at least 14 days and cleared by a licensed physician.

INTERVENTION IN THE WORK PLACE



RESPIRATORY HYGIENE AND COUGH ETIQUETTE

Tissues and alcohol hand rubs should be available in strategic places throughout the area/space. All toilet facilities should have adequate water and soap for handwashing.

PROMOTE MENTAL HEALTH

Mental and psychosocial support, such as but not limited to mindfulness activities/sessions, in-house counseling sessions, online counseling, and support groups, should be made available, as applicable, to the stakeholders of the sector (employees, students, etc.)

Employers must promote work-life balance through proper scheduling of activities and rotation of the workforce

REDUCE EXPOSURE OF VULNERABLE INDIVIDUALS

Households may institute daily monitoring of individuals at risk; and create contingency plans for accessing healthcare or purchasing of medication from pharmacy in case of emergency

Alternative work arrangements for vulnerable groups should be made when possible.

Specific lanes, areas, and/or facilities for the elderly, individuals with underlying conditions, and pregnant women should be designated to limit their exposure to others.

REFERENCES:

BOSH MANUAL – DOLE OSH CENTER
A GUIDE TO THE NEW NORMAL – HEALTH PROMOTION AND COMMUNICATION SERVICE DOH

ASSESSMENT:

MODULDE 7
OCCUPATIONAL HEALTH AND SAFETY AT WORK

Name of Student: ________________________________ Module No: 7

Course & Section: ________________________________ Course Title: **OJT ACADEMY**

In the past years, you have already experience working/doing an internship program. Try to remember your experience during your past internship program and answer the following question.

1. What types of accidents happen in your company/workplace (if this is your first time to take internship program, think about your usual surroundings)?

2. Do these accidents occur repeatedly?

3. Are accidents reported? State reasons for action taken

4. What actions did your company/workplace take in order to prevent the recurrence of similar accidents in the future?

5. Before you start your internship program before, did your company/workplace conducted a safety orientation program in the workplace?

Do you think it is important? Why?